SWIFT SHIP TRACKER

VIVEKANANDHA COLLEGE OF TECHNOLOGY FOR WOMEN [6130]

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SWIFT SHIP TRACKER

User Story:

The SwiftShipTracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. INTRODUCTION

1.1 Project Overview :

The SwiftShipTracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose :

The purpose of SwiftShipTracker is to:

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.

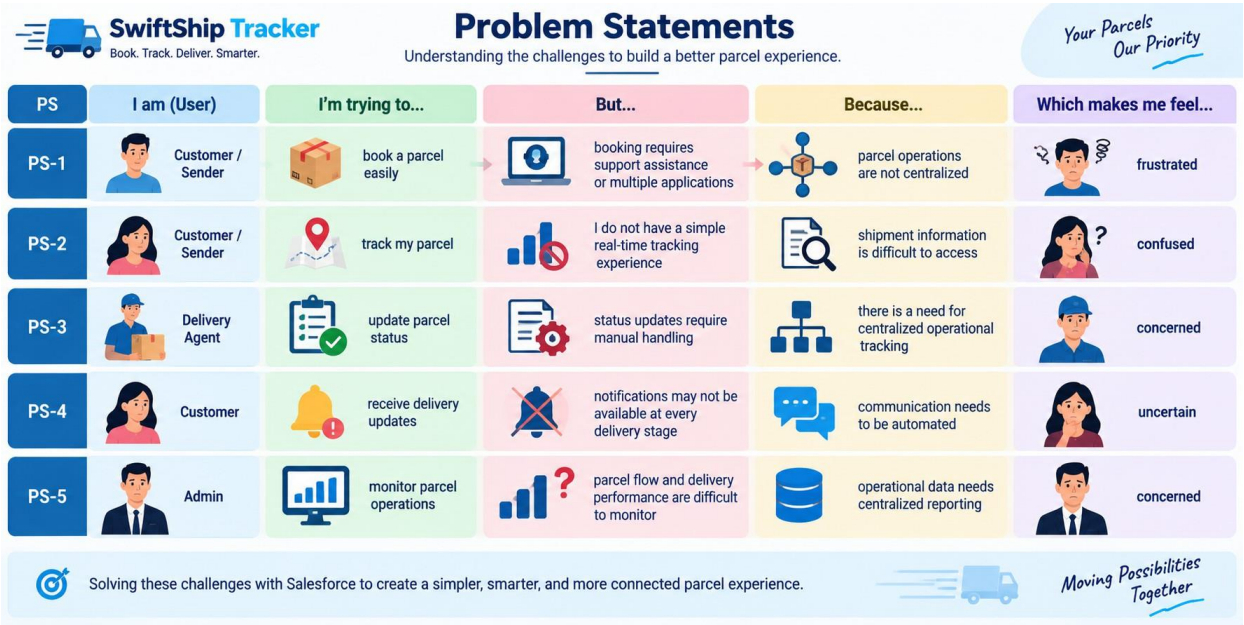
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.
- Improve agent efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access to customers, agents, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	book a parcel easily	booking requires support assistance or multiple applications	parcel operations are not centralized	frustrated
PS-2	Customer / Sender	track my parcel	I do not have a simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	update parcel status	status updates require manual handling	there is a need for centralized operational tracking	concerned
PS-4	Customer	receive delivery updates	notifications may not be available at every delivery stage	communication needs to be automated	uncertain

PS-5	Admin	monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

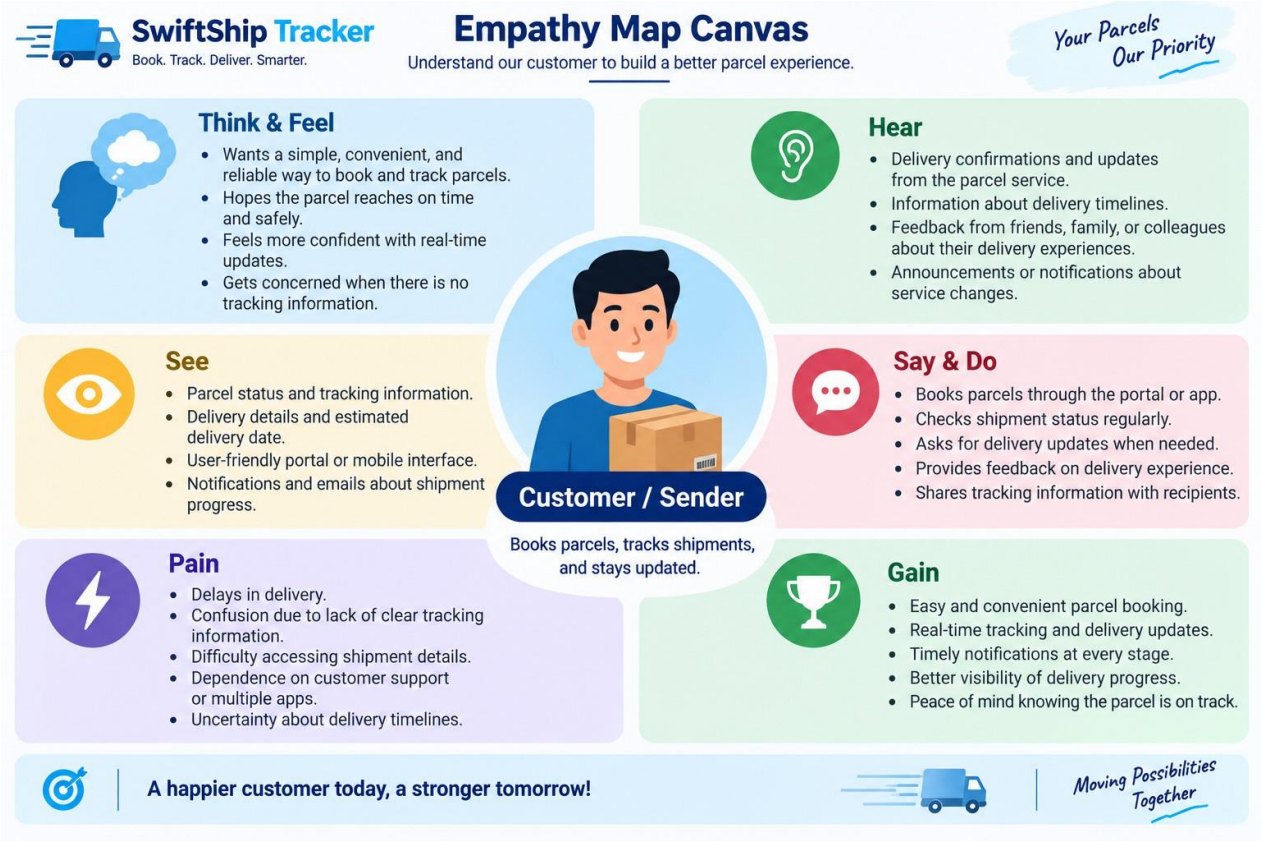


2.2 Empathy Map Canvas

User Persona: Customer / Sender

- **Think & Feel:** Wants a simple way to book and track parcels without repeatedly contacting support.
- **Hear:** Receives delivery information and status updates from the parcel service.
- **See:** Parcel details, tracking status, estimated delivery date, and delivery information.
- **Say & Do:** Books parcels, checks tracking information, and asks for delivery updates.
- **Pain:** Delays, confusion, lack of easy tracking, and dependence on support or multiple apps.

- **Gain:** Real-time visibility, self-service tracking, convenient booking, and timely notifications.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

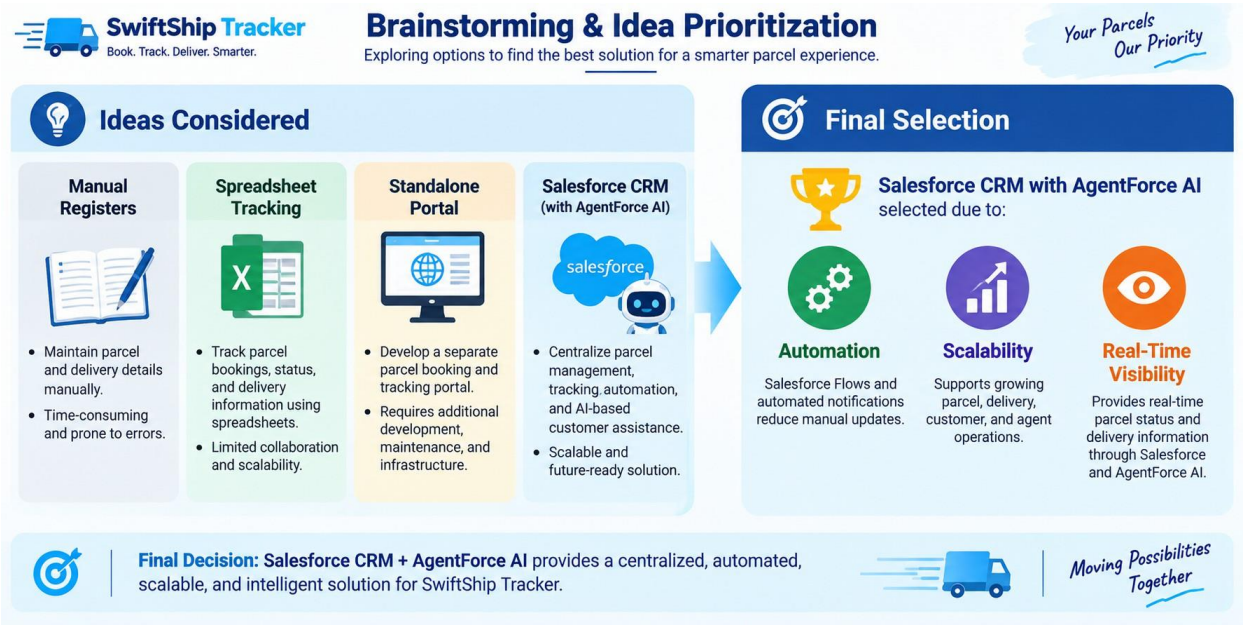
- Manual registers
- Spreadsheet tracking
- Standalone parcel portal
- Salesforce CRM with AgentForce AI

Final Selection

Salesforce CRM selected due to:

- Centralized parcel management

- Automation through Flows
- Real-time tracking visibility
- Conversational AI assistance
- Secure role-based access
- Reports and dashboards



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Parcel Booking
- Parcel Record Creation
- Dispatch & Shipment
- In-Transit Tracking
- Out for Delivery
- Delivery Confirmation
- Customer Notification
- Feedback / Support



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Parcel booking and record creation
FR-2	Parcel status tracking
FR-3	Weight management
FR-4	Estimated delivery date management
FR-5	Sender information management
FR-6	Receiver information management
FR-7	Delivery and current-location tracking

FR-8	Agent parcel status updates
FR-9	Automated customer notifications
FR-10	AI-powered parcel tracking
FR-11	Reports and dashboards
FR-12	Customer access through Experience Cloud

Non-Functional Requirements

NFR	**Description**
Usability	Intuitive Salesforce and customer-facing interface
Security	Role-based and field-level access control
Reliability	Accurate parcel status and delivery information
Performance	Timely Flow and AgentForce responses
Availability	Salesforce cloud-based access
Scalability	Supports growing parcel and delivery operations
Data Integrity	Controlled access and relationship-based parcel data
Maintainability	Flows, prompts, topics, and configurations can be updated

3.3 Data Flow Diagram

Flow:

Customer / Sender

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Parcel Booking

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Parcel Record

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Sender + Receiver Information

↓

Delivery Record

↓

Parcel Status Update

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Current Location & Estimated Delivery

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Automated Notifications

↓

Customer Tracking / AgentForce AI

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Reports & Dashboards

AI Tracking Flow

Customer Query

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AgentForce AI

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Prompt Builder – Retrieve Parcel Details

↓

Parcel Record

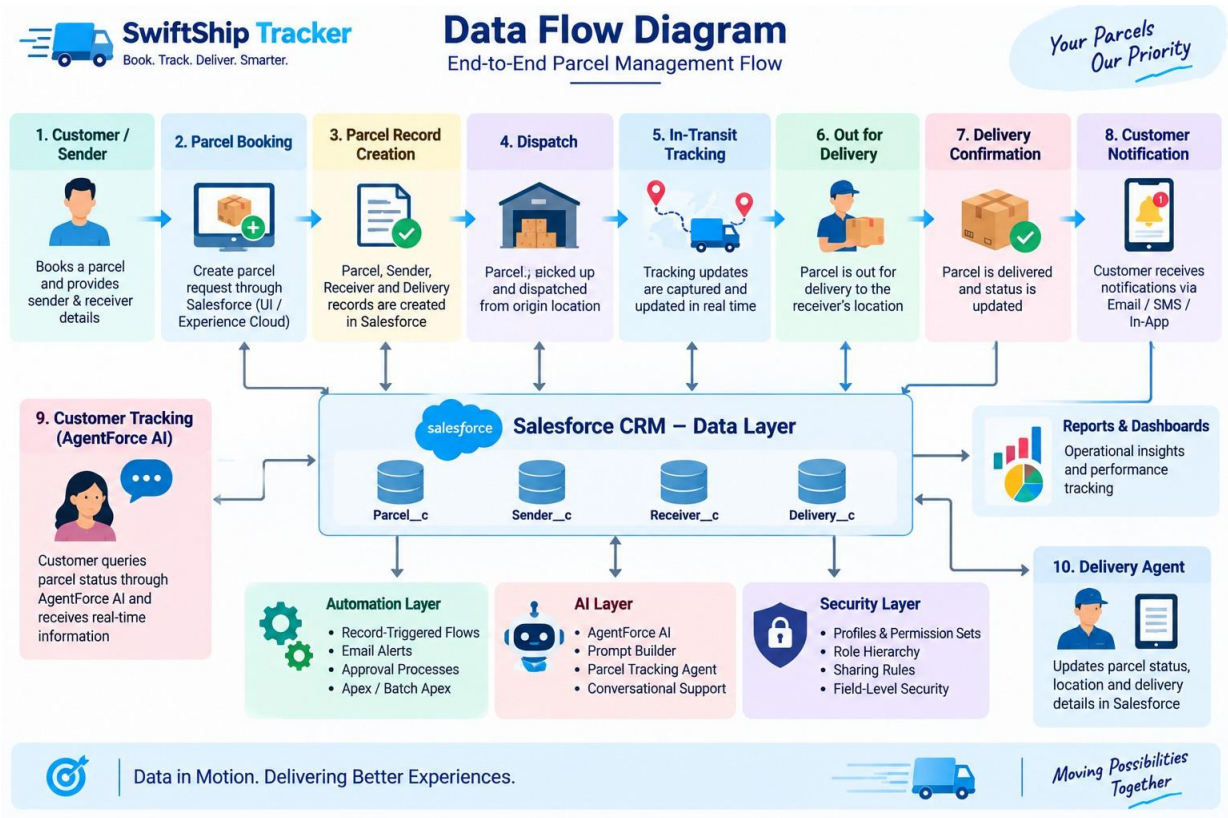
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Status / Weight / Estimated Delivery

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Conversational Response

The Prompt Builder implementation retrieves Parcel ID, Status, Weight, and Estimated Delivery Date information for tracking responses.



3.4 Technology Stack

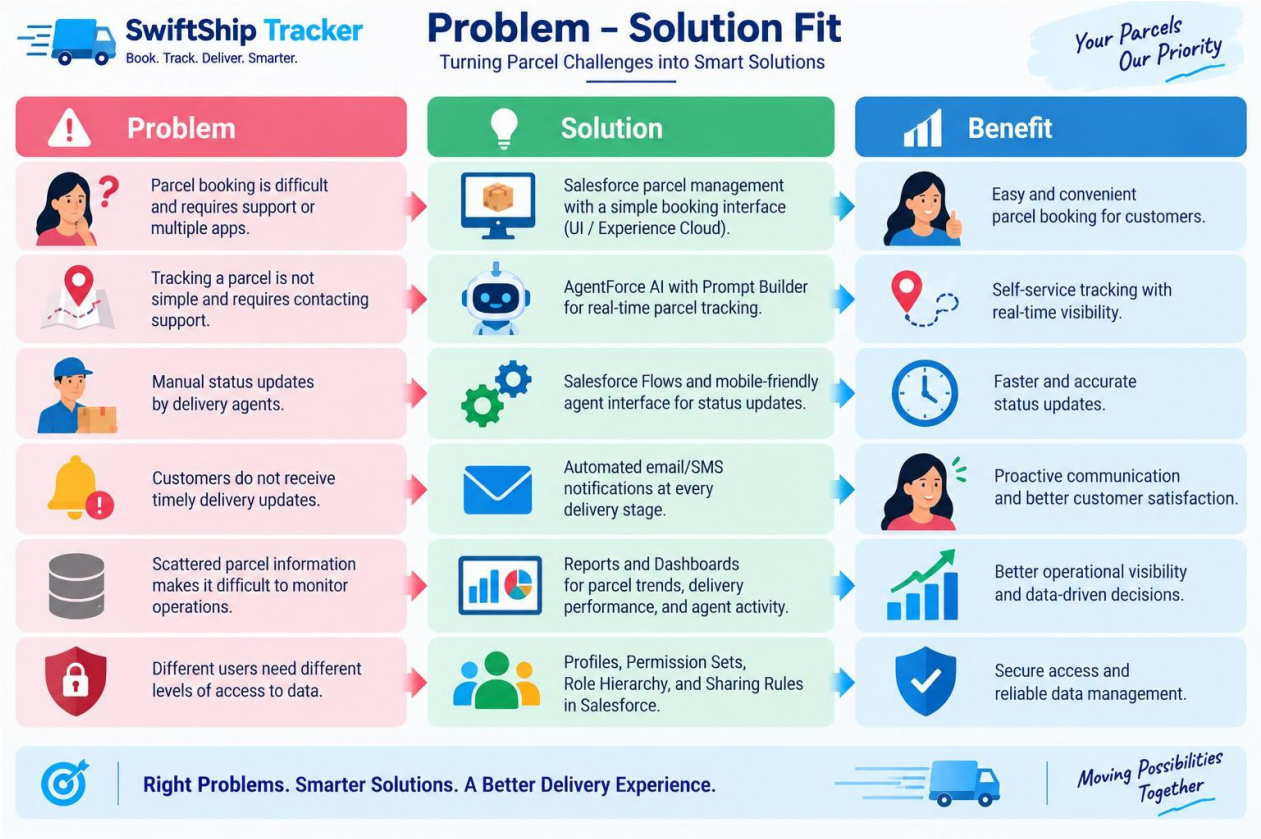
Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience Cloud
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates
Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Parcel booking is difficult	Salesforce Parcel management	Centralized booking
Tracking requires support	AgentForce AI + Prompt Builder	Self-service tracking
Manual status updates	Salesforce Flows	Automated processes

Problem	Solution	Benefit
Delivery communication gaps	Email Alerts	Timely notifications
Scattered shipment information	Parcel, Delivery, Sender & Receiver objects	Structured data
Difficult operational monitoring	Reports & Dashboards	Better visibility
Different users need different access	Profiles, Roles & Permission Sets	Secure access



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

4.3 Solution Architecture

- Users
- Customer | Delivery Agent | Support Staff | Admin

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- Salesforce Lightning / Experience Cloud

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- SwiftShipTracker Salesforce Platform
- Data Layer
  - Parcel
  - Delivery
  - Sender
  - Receiver
- Automation Layer
  - Flows
  - Approval Processes
  - Apex Classes

    - Batch Apex
    - Email Alerts
- **AI Layer**
    - AgentForce AI
    - Prompt Builder
    - Parcel Tracking Agent
- Security Layer
    - Profiles
    - Roles
    - Permission Sets
    - Sharing Rules
    - Field-Level Security

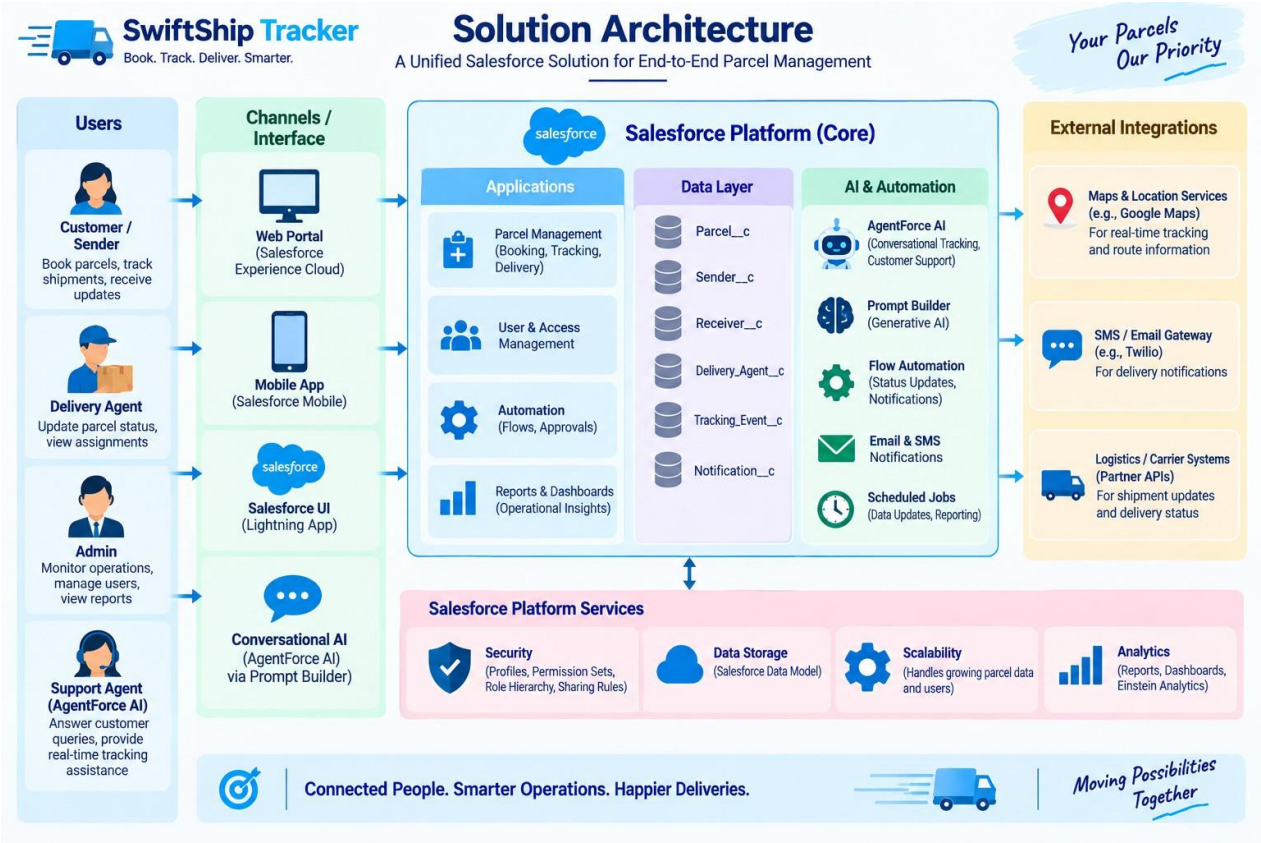
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- Reports & Dashboards

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- Parcel Tracking & Operational Insights

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender, and Receiver objects with proper relationships so that parcel and delivery data is stored and managed systematically.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily navigate and manage parcel and delivery operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate parcel details, mandatory fields, and booking inputs so that invalid or incomplete parcel records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a delivery coordinator, I want automated Flows to update parcel status and delivery information so that parcel processing requires minimal manual effort.	3	High	Member4
Sprint-4	Apex	USN-6	As a system, I want Apex triggers and classes to automatically perform critical parcel and delivery actions so that important processes are handled accurately.	5	High	Member1
Sprint-4	Apex	USN-7	As an inventory/operations manager, I want Batch Apex jobs to process and update parcel or delivery records automatically so that operational data remains accurate and up to date.	5	High	Member2
Sprint-5	Reports	USN-8	As a manager, I want reports showing parcel status, delivery performance, and operational metrics so that I can monitor SwiftShip operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards & AI	USN-9	As a manager, I want dashboards showing parcel status, delivery performance, and key metrics, while Agentforce provides AI-powered tracking assistance, so that I can monitor operations efficiently.	4	Medium	Member4



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShipTracker parcel management system.	3	High	Member1

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		
Sprint-5	20	6 Days	21 Nov 2022	26 Nov 2022		
Sprint-6	20	6 Days	28 Nov 2022	03 Dec 2022		

6. Project Development Phase:

Phase 1: Requirement Analysis & Planning

Objective:

“Customers often face delays and confusion while booking, tracking, and managing parcel deliveries because they need to call support or use multiple apps. A conversational AI agent can simplify this by allowing customers to book parcels, get live tracking, and manage deliveries in one place.”

Key Requirements:

• Requirement gathering from end users and delivery agents.

• Defining system scope and user personas (Customer, Agent, Admin).

• Designing the data model with relationships between Parcel, Sender, Receiver, and Delivery.

• Identifying use cases such as parcel booking, tracking, and delivery update automation.

• Selecting tools: Salesforce Flows, Prompt Builder, and AgentForce AI.

Defining Project Scope & Objectives

Project Scope:

1. Build a scalable CRM solution for parcel delivery businesses to manage customer bookings, shipments, and delivery operations efficiently.
2. Support the end-to-end parcel lifecycle — from parcel booking → dispatch → tracking → delivery confirmation.
3. Include comprehensive data model design, automation (Flows, Apex, Email Alerts), AI-powered prompt builder, and reporting dashboards for effective management.
4. Provide role-based access for admins, delivery agents, and customers through Salesforce profiles, permission sets, and Experience Cloud.
5. Ensure real-time parcel visibility for customers and simplified operational tracking for agents through AI-integrated conversational support.

Objectives Summary:

• Streamline parcel operations by automating booking, tracking, and delivery notifications to reduce manual updates.

• Improve customer satisfaction with transparent, real-time shipment status and self-service tracking options.

• Enhance agent efficiency through centralized dashboards and guided Flows for quick status updates.

• Enable business insights with reports and dashboards showing parcel trends, delivery performance, and agent activity.

• Ensure security and reliability by implementing Salesforce role hierarchies, sharing rules, and audit-based data access.

Gathering & Analysing User Needs

Users Involved:

- Customers / Senders – End users who book parcels, track deliveries, and receive real-time updates through the portal or AI agent.
- Delivery Agents – Manage assigned parcels, update delivery status, and ensure timely handover to customers.
- Customer Support Team – Handle delivery-related issues, respond to customer queries, and ensure smooth communication.

- Admin Team – Configure the Salesforce environment, manage user roles, security settings, and monitor operational performance.
- System Integrator / Developer – Responsible for implementing automation, AI agent setup, and backend integrations.

Key Functional Needs:

- Seamless parcel booking and tracking interface for customers through Salesforce Experience Site or agent assistance.
- Ability for agents to update parcel status (Booked → In Transit → Out for Delivery → Delivered) in real time.
- Automated notifications and email alerts to customers on every delivery stage update.
- Support for Sender and Receiver details management with address, contact, and email fields for communication.
- AI Agent integration (via Prompt Builder) for real-time parcel status queries and automated assistance.
- Reports and dashboards to monitor parcel flow, delivery performance, and agent activity for admins.
- Security controls with role-based access, profiles, and permission sets for different user levels.

Tools Used:

- Salesforce Developer Org – For configuration, development, and testing.
- Prompt Builder (AgentForce AI) – To create intelligent parcel tracking prompts and responses.
- Salesforce Flow Builder – To automate booking, tracking, and delivery updates.
- Google Sheets / Docs – For initial data planning, requirement documentation, and system mapping.

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned:

- Custom Objects:

Parcel__c, Delivery__c, Sender__c, Receiver__c

- Standard Objects:

Account, Contact, Case, Task, EmailMessage, User

- Automations:

Record-Triggered Flows, Auto-Launched Flows, Approval Processes, Scheduled Flows

- Apex: Apex Classes for automated parcel operations (e.g., Lead-to-Parcel mapping, delivery notifications), Batch Apex for periodic updates or notifications

- UI: Lightning App Page (SwiftShipTracker App), Dynamic Forms for Parcel Details, Lightning Record Pages, and Experience Cloud for Customer Access

- AgentForce AI & Prompt Builder: Integrated AI agent trained to fetch parcel data based on Parcel ID and respond to tracking or delivery status queries

- Email Services: Custom Email Alerts and Templates for booking confirmations, delivery updates, and completion acknowledgments

- Security: Profiles, Permission Sets, Role Hierarchies, Field-Level Security, and Sharing Rules to ensure controlled access for Admin, Agent, and Customer roles

Designing Data Model and Security Model

Data Model Includes:

- Parcel__c (Custom Object)
- Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date.
- Linked to Sender__c and Receiver__c objects to fetch sender and receiver details.
- Connected with Delivery__c object to track real-time parcel location and status.
- Delivery__c (Custom Object)
- Stores delivery-specific information such as current location, estimated delivery date, and linked parcel.
- Connected with Sender__c and Parcel__c for relationship-based tracking.
- Used by agents to update and monitor delivery status during the parcel lifecycle.
- Sender__c (Custom Object)
- Captures sender-related information such as address, contact number, and email.
- Linked to multiple parcels to track sender history and associated deliveries.
- Used for sending booking confirmations and delivery updates via automated emails.
- Receiver__c (Custom Object)

- Stores receiver details such as address, contact, and email for communication and delivery confirmation.
- Linked to Sender__c and Parcel__c to associate both ends of the shipment process.
- Used by agents for delivery reference and automated email notifications upon completion.

Security Model Design:

- Role Hierarchy:

Administrator → Delivery Manager → Agent → Customer

- Profiles:

Admin Profile: Full access to all objects and records.

Delivery Agent Profile: Read/Edit access to Parcel, Delivery, Sender, and Receiver objects.

Customer Profile: Read-only access to their own parcel details via Experience Cloud.

Support Staff Profile: Access to Cases and Parcel details for issue resolution.

- Permission Sets:

Used to grant additional privileges (e.g., “Update Delivery Status,” “Access Agent Console”).

- Record-Level Security:

Sharing Rules applied for Parcel__c and Delivery__c to allow controlled access for agents and managers.

Customers can only view records linked to their own Sender or Receiver details.

- Field-Level Security:

Sensitive fields like Sender Email, Receiver Contact, and Location Coordinates restricted to Admin and Agent profiles only.

Phase 2: Salesforce Development –Backend & Configurations

This phase focused on developing the core logic of SwiftShipTracker Custom objects were created for Parcel, Sender, Receiver, and Delivery with relationships. Automations were built using Flows for parcel status updates and approval processes.

Milestone 1- Salesforce Account

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

What Is Salesforce?

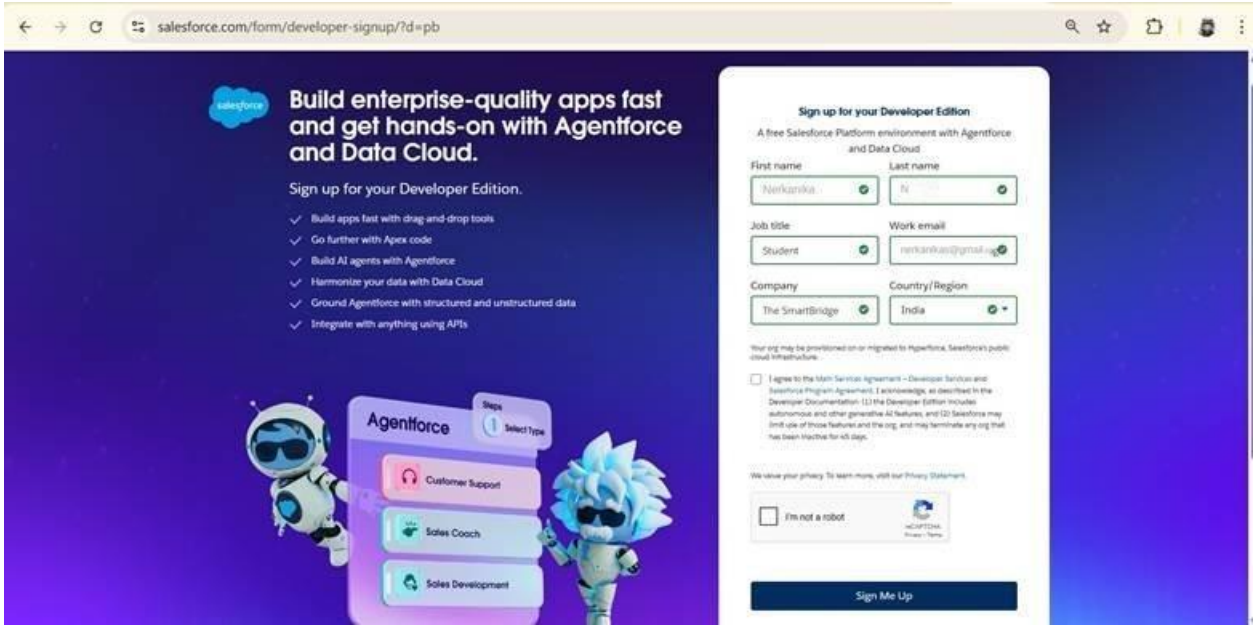
Salesforce is your customer success platform, designed to help you sell, service, market, analyse, and connect with your customers.

Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup

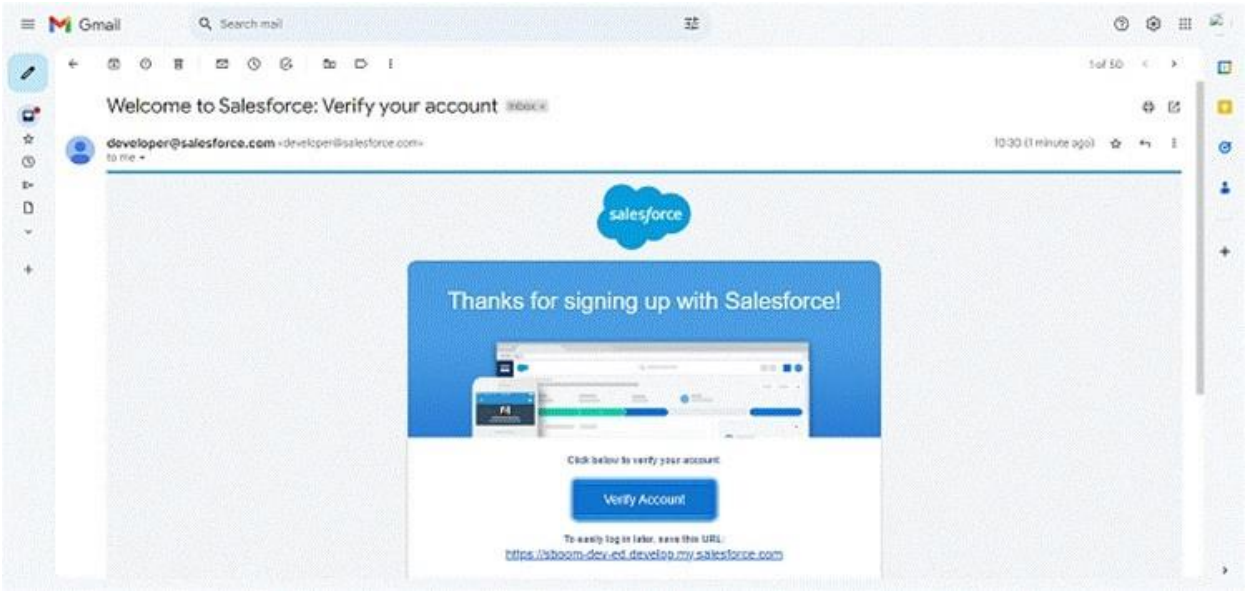
2. On the sign-up form, enter the following details:



1) First name & Last name
   2) Email
   3) Job Title: Developer
   4) Company: College Name
   5) Country: India

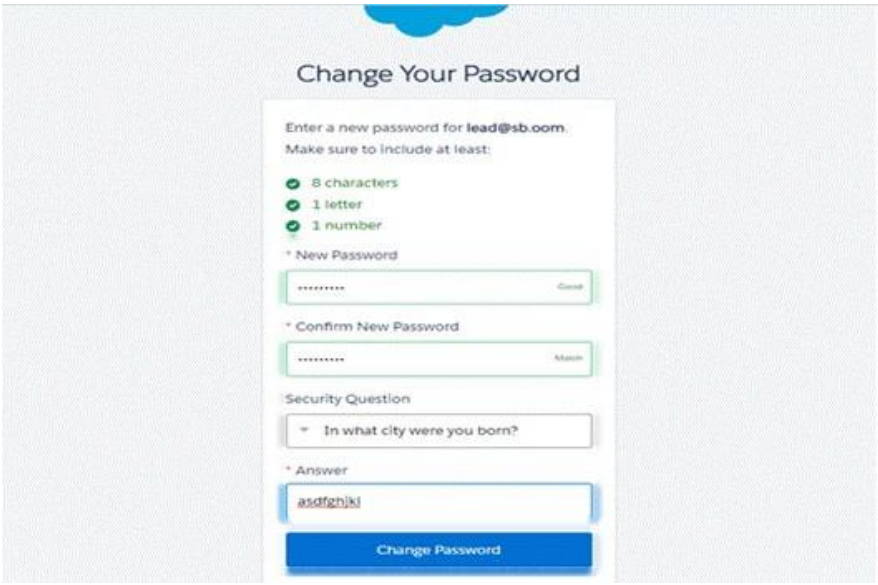
Click on sign me up after filling these.

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30mins and sometimes 2 hours.

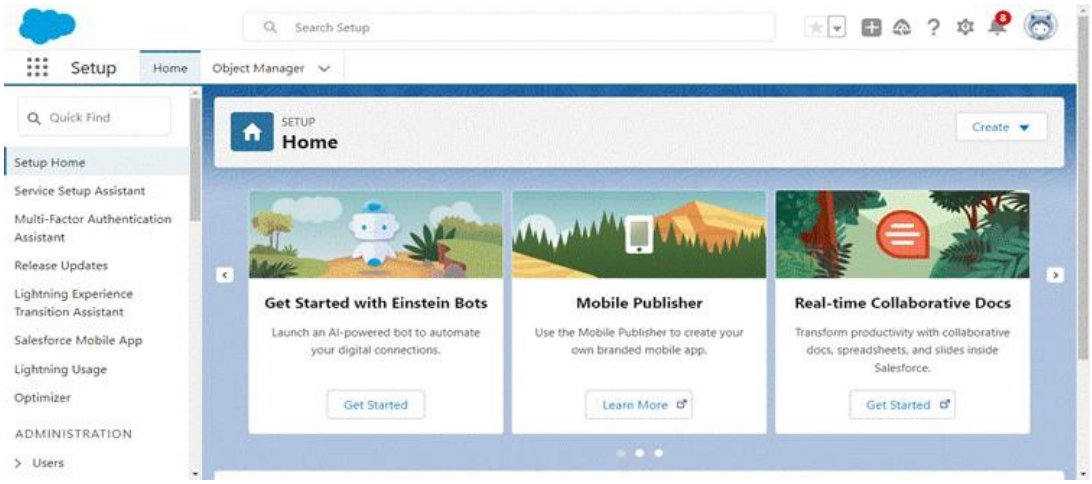


2. Click on Verify Account

3. Give a password and answer a security question and click on change password.



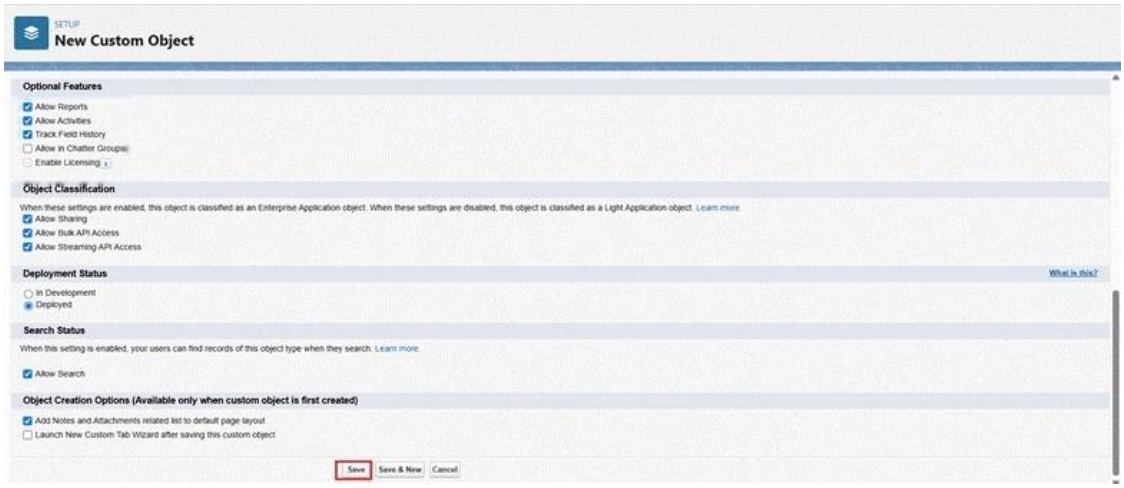
4. Then you will redirect to your salesforce setup page.

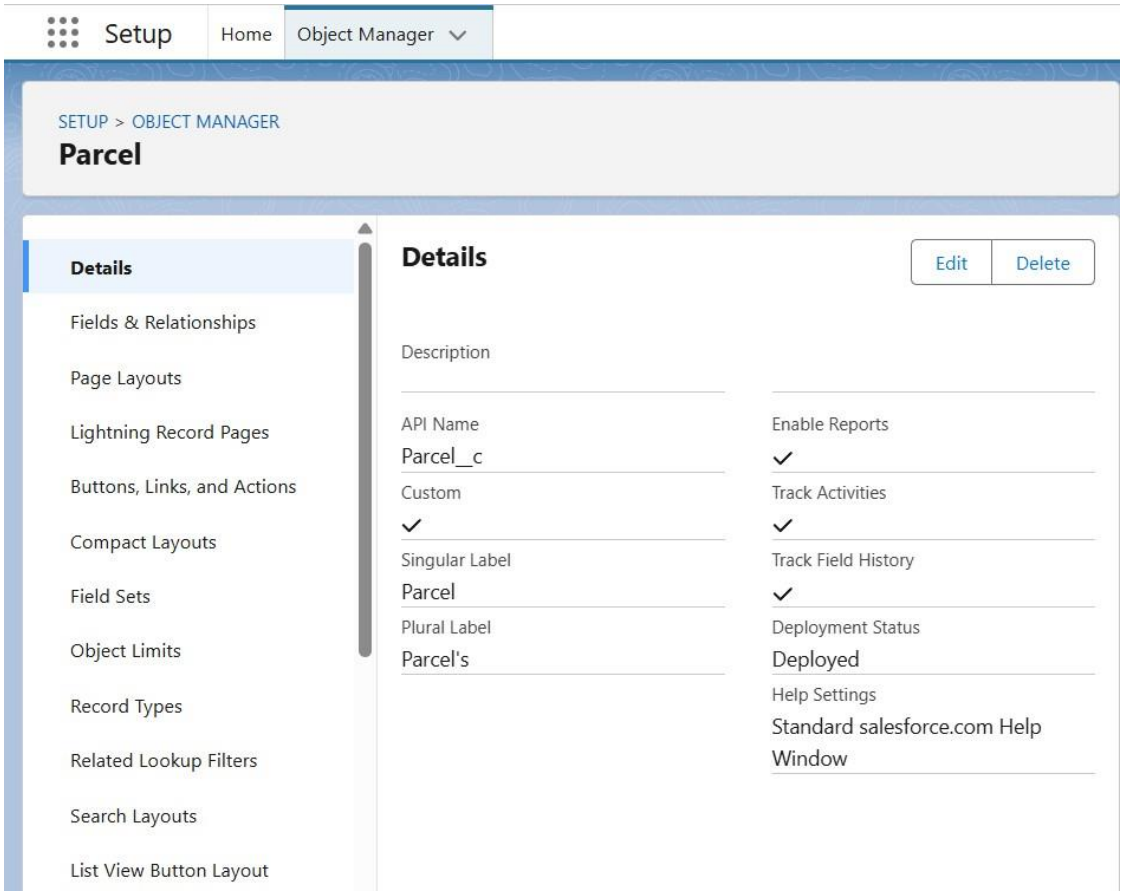


Milestone 2- Objects Creation

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Parcel
5. Enter Plural label name as Parcels
6. Enter Record Name as Parcel Name
7. Select Data Type as Text: Parcel Name
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New





Creating Delivery Object

To create an object:

1. From the setup page

2. Click on Object Manager

3. Click on Create >> Click on Custom Object.

4. Enter the label name as Delivery

5. Enter Plural label name as Deliveries

6. Enter Record Name as DeliveryName

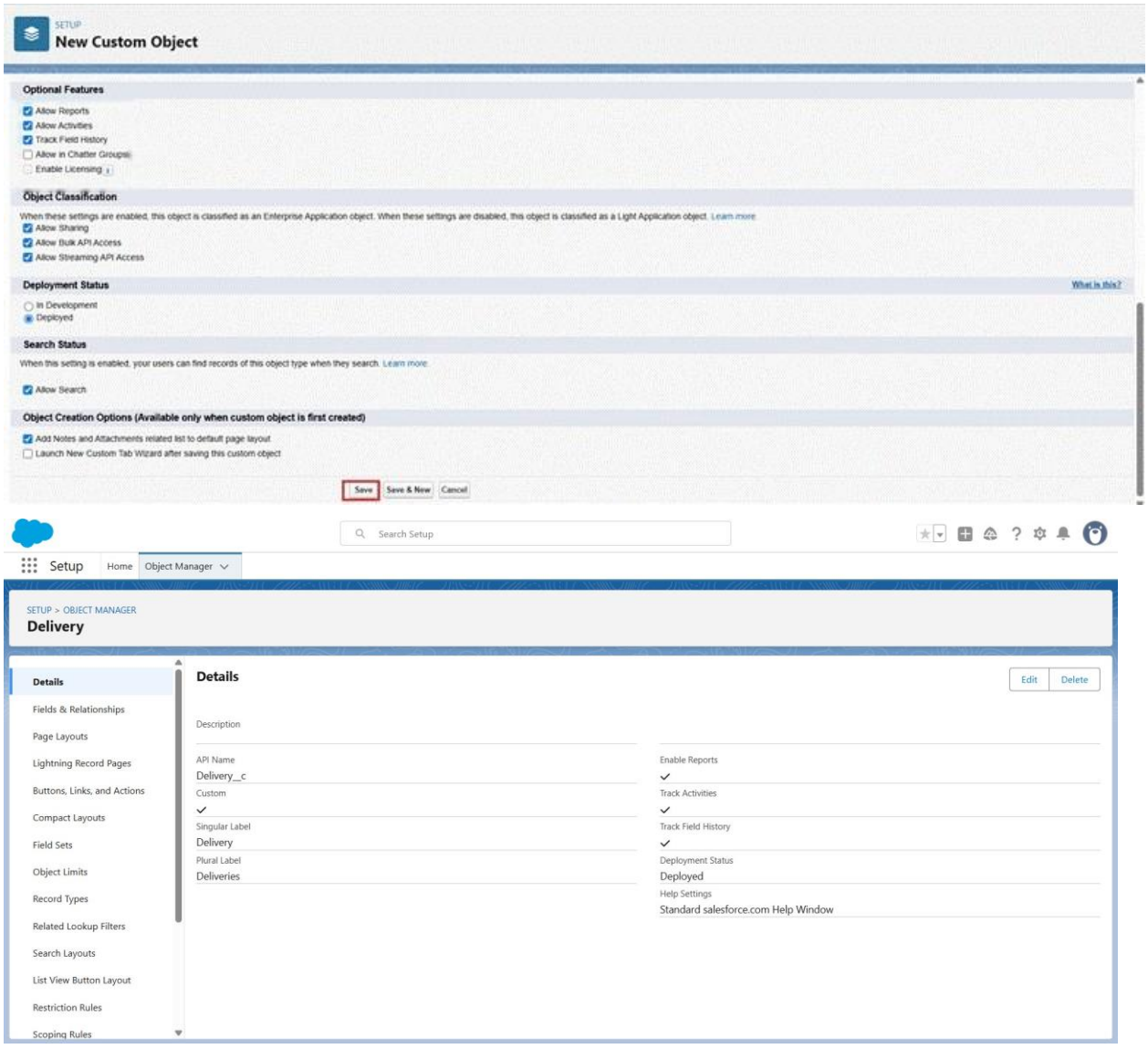
7. Select Data Type as Text: DeliveryName

8. Select Allow reports.

9. Select Allow search.

10. Allow Track Field History

11. Click on Save and New



Creating Sender Object

To create an object:

1. From the setup page

2. Click on Object Manager

3. Click on Create >> Click on Custom Object.

4. Enter the label name as Sender

5. Enter Plural label name as Sender's

6. Enter Record Name as SenderName

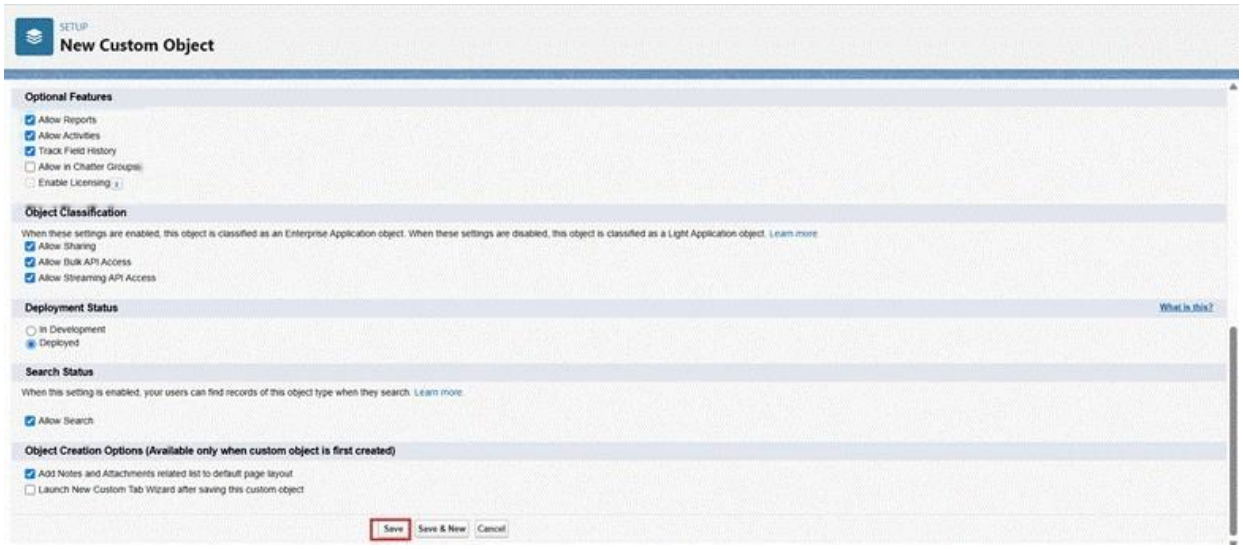
7. Select Data Type as Text: SenderName

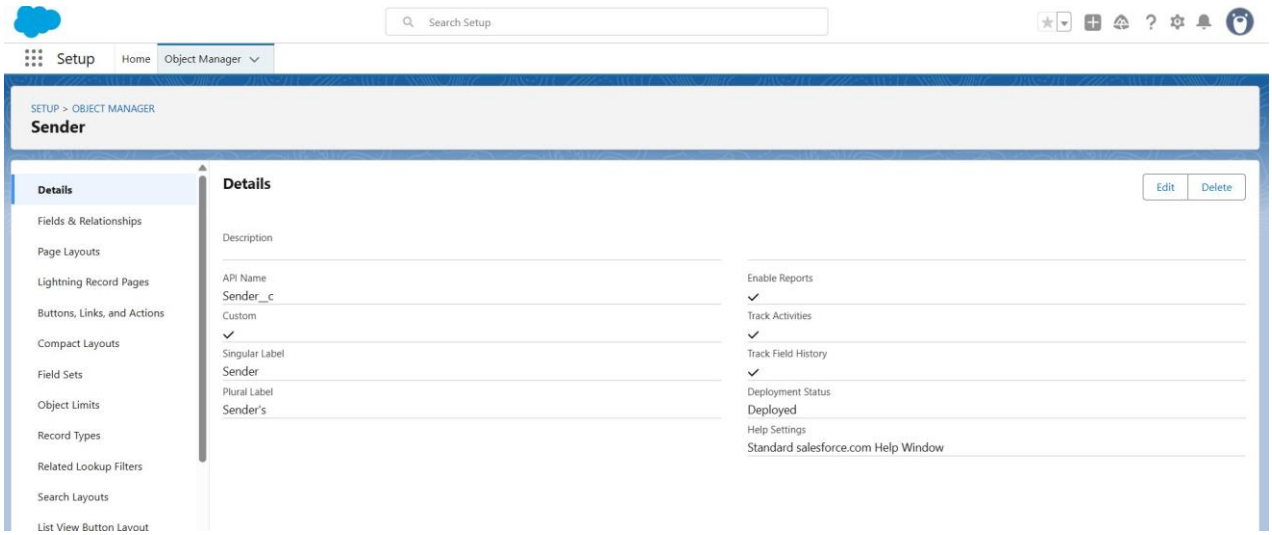
8. Select Allow reports.

9. Select Allow search.

10. Allow Track Field History

11. Click on Save and New





Creating Receiver Object

To create an object:

1. From the setup page

2. Click on Object Manager

3. Click on Create >> Click on Custom Object.

4. Enter the label name as Receiver

5. Enter Plural label name as Receiver's

6. Enter Record Name as ReceiverName

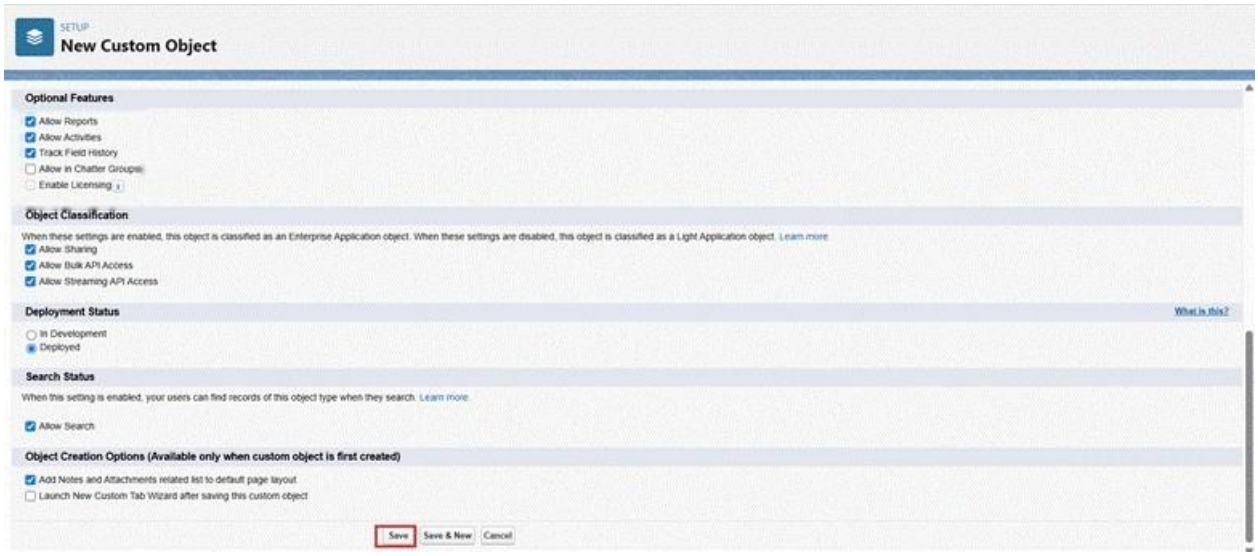
7. Select Data Type as Text: ReceiverName

8. Select Allow reports.

9. Select Allow search.

10. Allow Track Field History

11. Click on Save and New



Milestone 3 - Tabs

Creating a tab for Parecel Object
1. Go to the setup page → type Tabs in Quick Find bar

2. Click on tabs

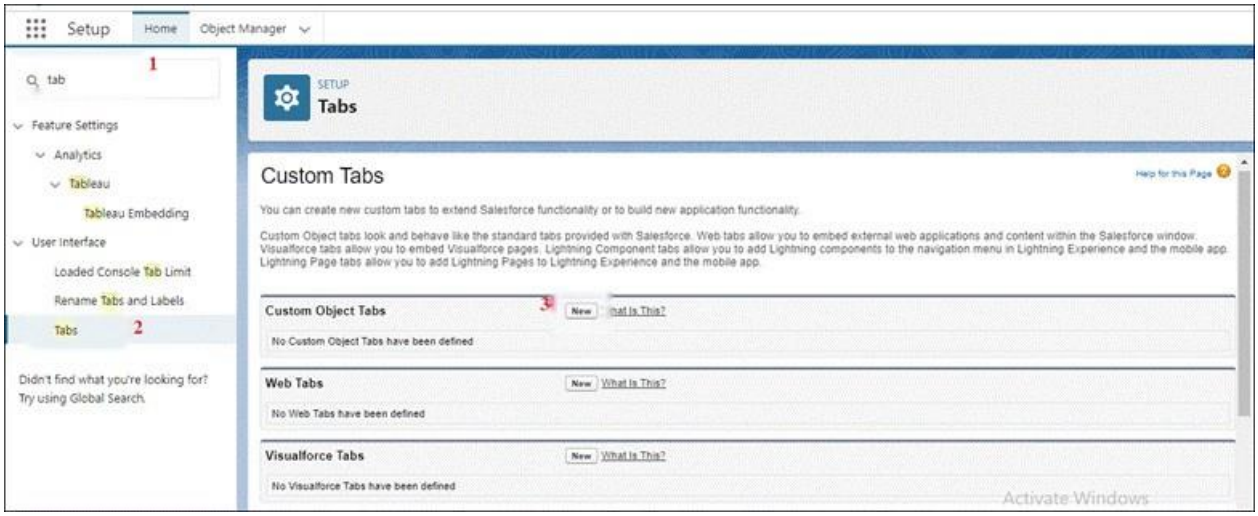
3. Click on New (under custom object tab).

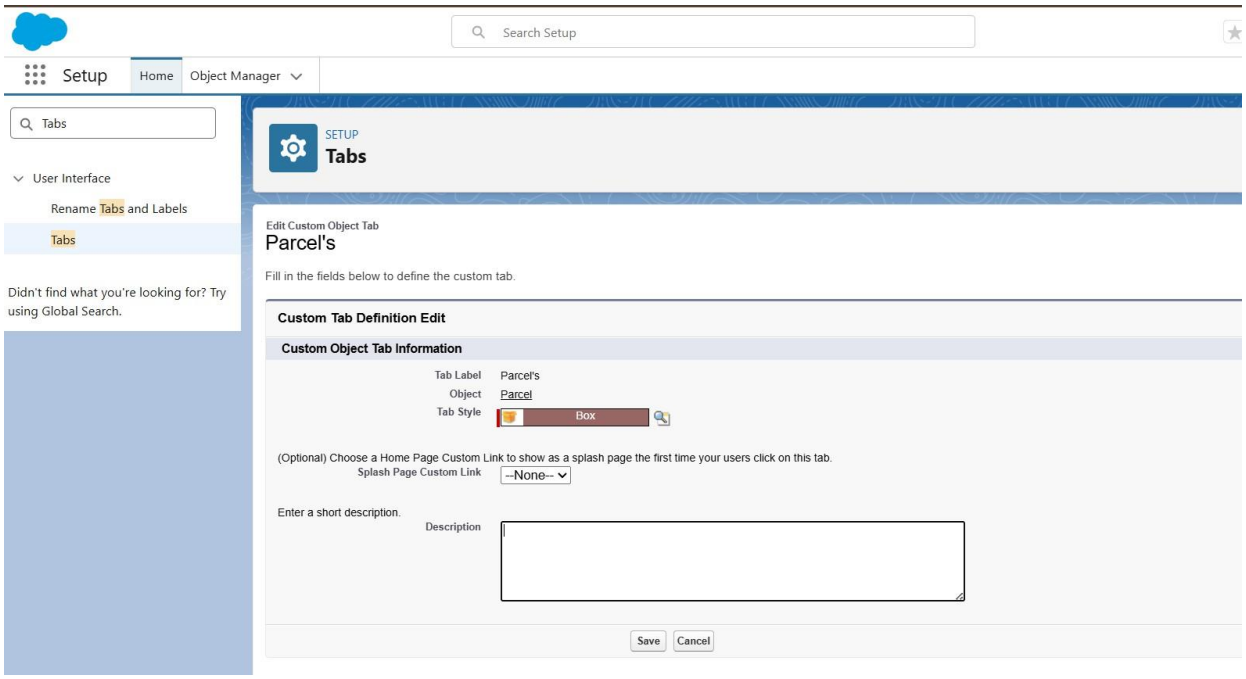
4. Select Object (Parcel) >> Select the tab style

5. Click on Next >> (Add to profiles page) keep it as default >> Click on Next (Add to Custom App) uncheck the include tab.

6. Make sure that the Append tab to the user's existing personal customizations is checked.

7. Click save

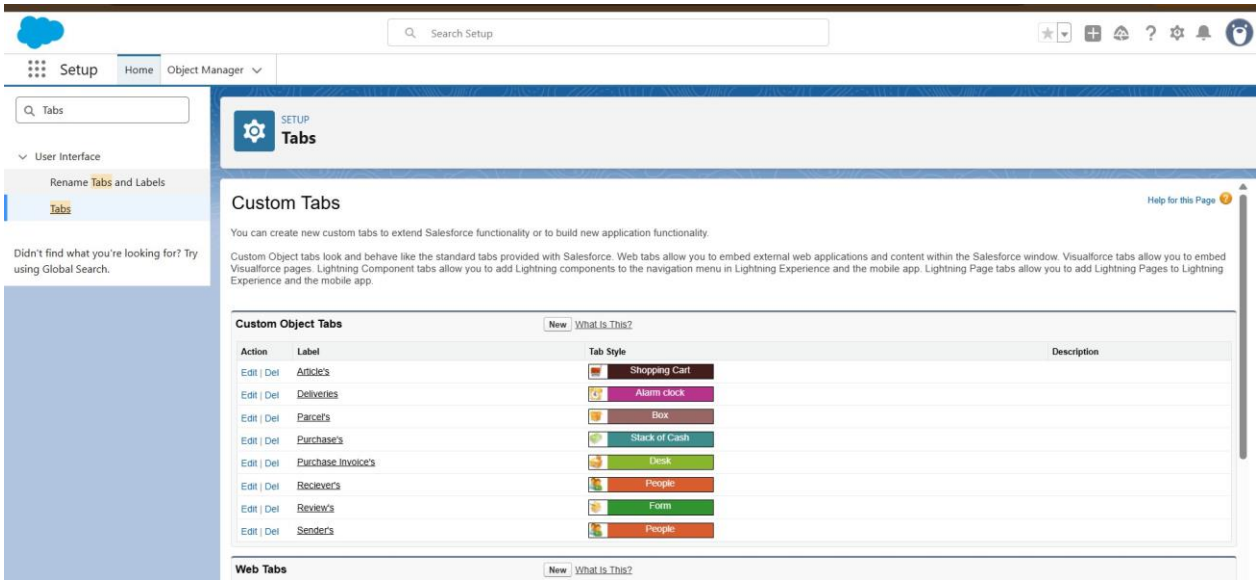




More Tab Creation

1. Go to Setup → type Tabs in the Quick Find box.

2. Click Tabs under the User Interface section.
3. Click New under Custom Object Tabs.
4. Select the object (Delivery, Sender, or Receiver) → choose a Tab Style.
5. Click Next → keep the Add to Profiles settings as default → click Next.
6. On the Add to Custom Apps page, uncheck “Include Tab.”
7. Ensure Append tab to the user’s existing personal customizations is checked.
8. Click Save and repeat the steps for each object (Delivery, Sender, Receiver).



Milestone 4- Fields & Relationships

Create fields and Relationships for the custom objects created in the previous milestone.

Creating Fields and Relationships for Parcel Object

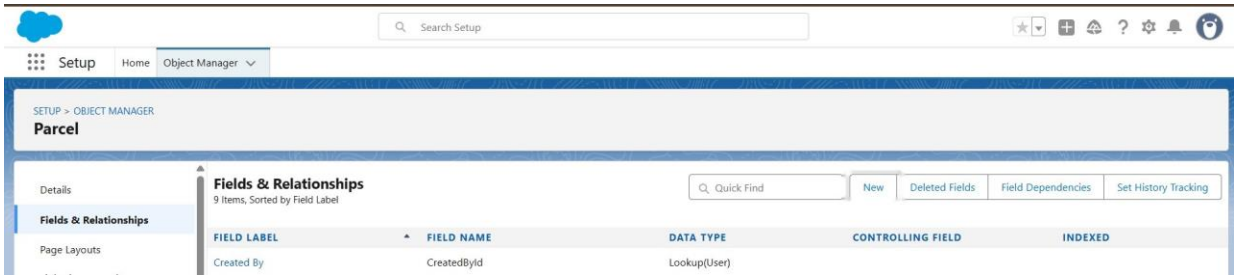
Beside the Quick Find box you have, Object Manager click on it.

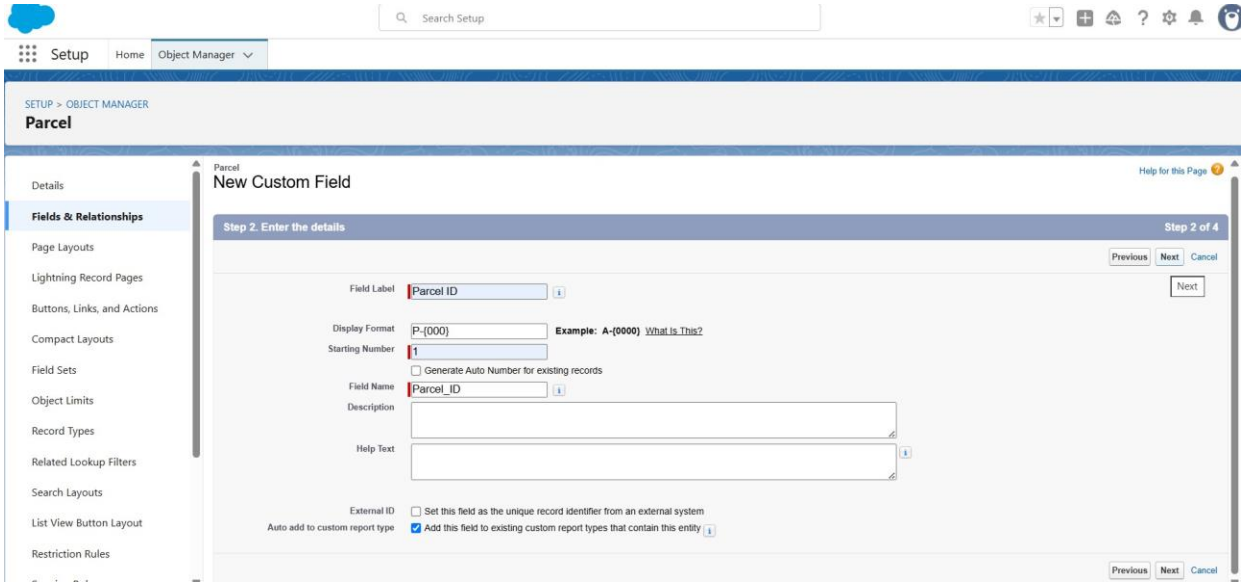
1. Search for and select your Parcel object.

Field 1: Parcel ID

- Go to Fields & Relationships → click New

  Enter:
- Data type== auto number→P-{000}





- Click Next → Next → Save & New.

3. Follow step 1 and 2 to create the remaining fields(Status, Weight, Estimated Delivery, Sender) of the Parcel object mentioned in the table below.)

1. Parcel Object

- **API Name:** Parcel__c
- **Purpose:** Store parcel information

Field Name	API Name	Data Type	Description
Parcel ID	Parcel_ID__c	Auto number	Unique ID for each parcel
Status	Status__c	Picklist	[Booked, In Transit, Out for Delivery, Delivered]
Weight	Weight__c	Number	Weight of Parcel
Estimated Delivery	Estimated_Delivery_Date__c	Date	Expected Delivery Date
Sender	Sender__c	Lookup Relationship	Gets Details from Sender

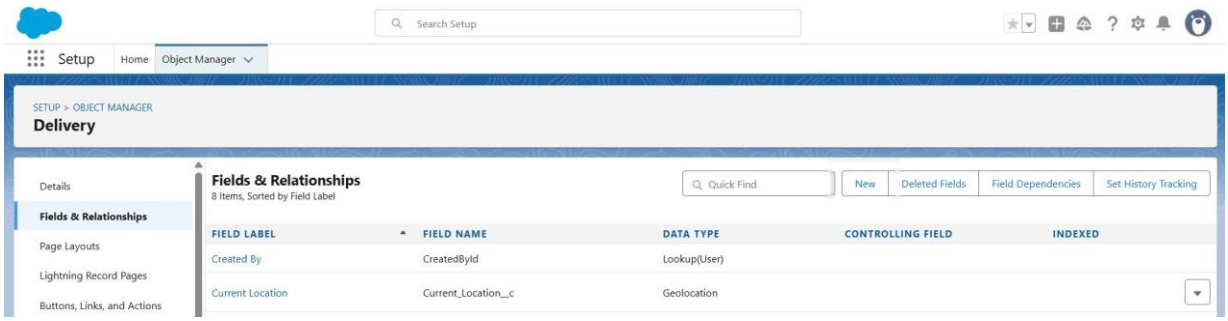
Creating Fields And Relationships For Delivery Object

Beside the Quick Find box you have, Object Manager click on it.

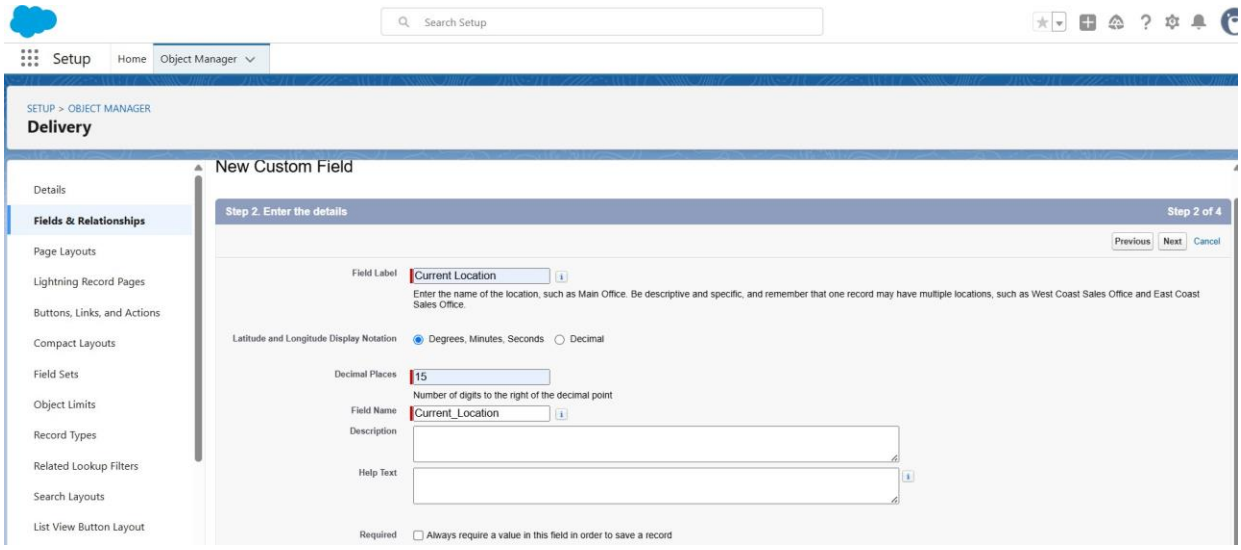
1. Search for and select your Delivery object.

Field 1: Current Location

- Go to Fields & Relationships → click New



- Choose Data Type: Geolocation → click Next.
  Field Label: Current Location
  Click Next → Next → Save & New.



- 3. Follow step 1 and 2 to create the remaining fields(Estimated Delivery, Sender, Parcel) of the Delivery object mentioned in the table below.

Field Name	API Name	Data Type	Description
Current Location	Current_Location__c	Geolocation	Location in terms of longitude and latitude.
Estimated Delivery	Estimated_Delivery_Date__c	Date	Expected Delivery Date
Sender	Sender__c	Lookup Relationship	Gets Details from Sender
Parcel	Parcel__c	Lookup Relationship	Gets Details from Parcel

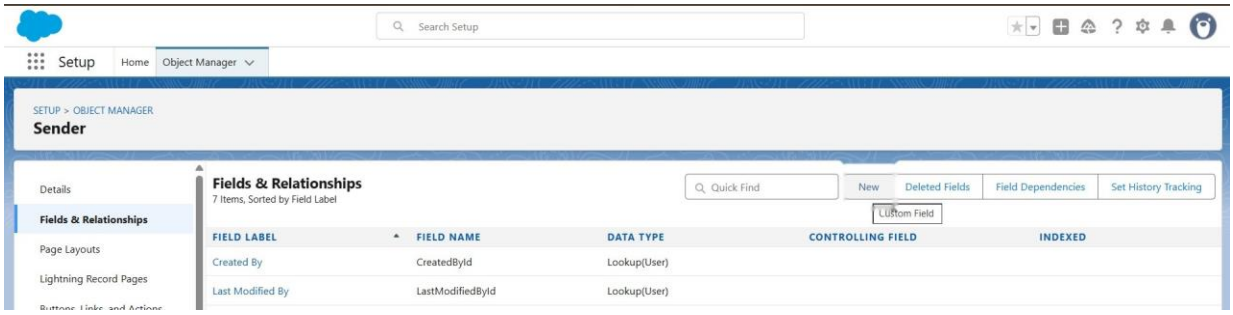
Creating Fields And Relationships For Sender Object

Beside the Quick Find box you have, Object Manager click on it.

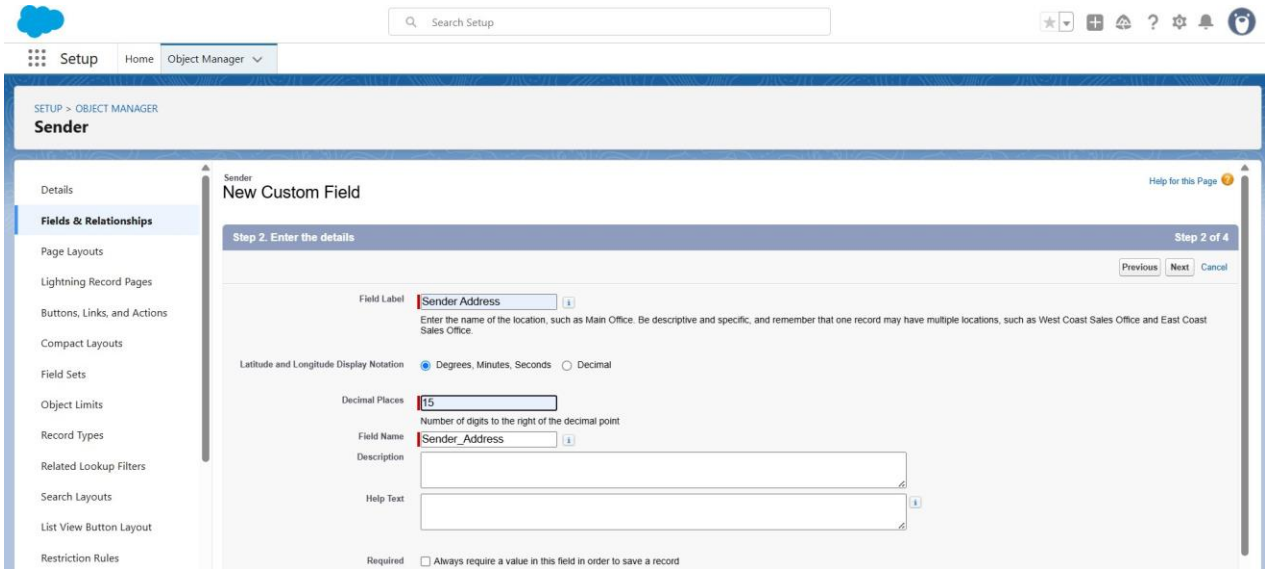
1. Search for and select your Sender object.

Field 1: Sender Address

- Go to Fields & Relationships → click New



- Choose Data Type: Geolocation → click Next.
  Field Label: Sender Address
  Click Next → Next → Save & New.



- 3. Follow step 1 and 2 to create the remaining fields (Sender Contact, Sender Email) of the Sender object mentioned in the table below.

Field Name	API Name	Data Type	Description
Sender Address	Sender_Adress__c	Geolocation	Location in terms of longitude and latitude.
Sender Contact	Sende_Contact__c	Phone	To contact the sender
Sender Email	Sender_Email__c	Email	To send email Notification.

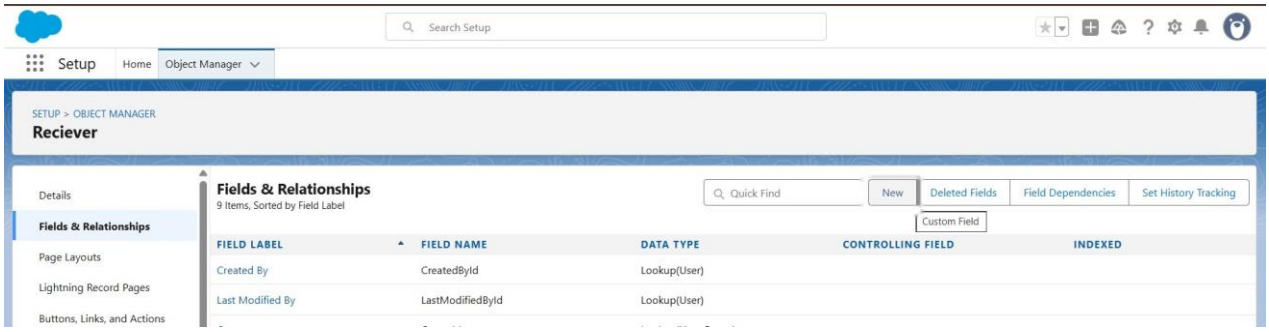
Creating Fields And Relationships For Receiver Object

Beside the Quick Find box you have, Object Manager click on it.

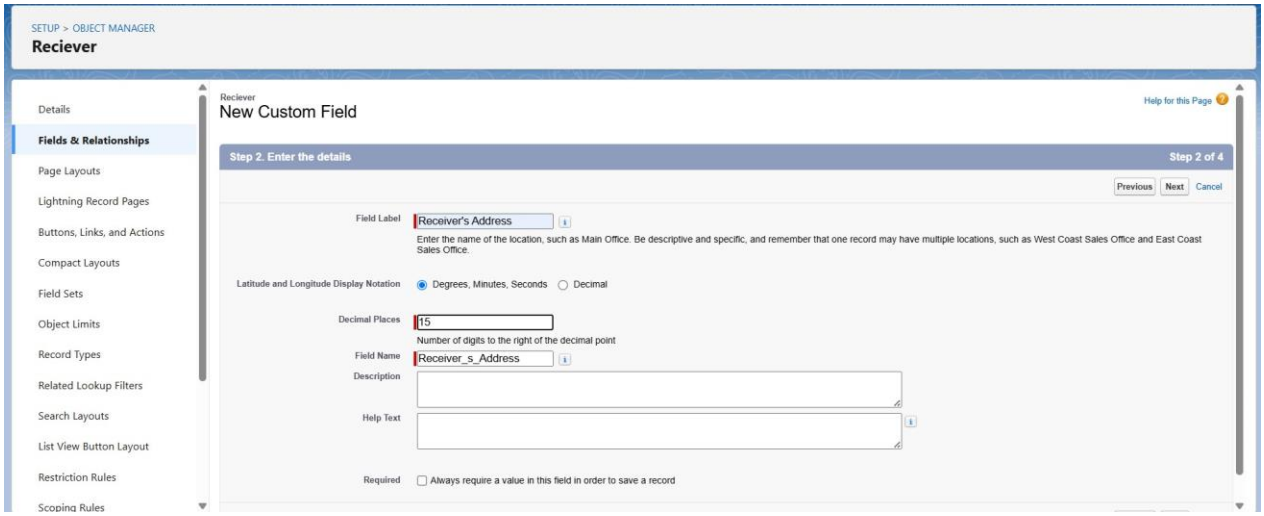
1. Search for and select your Receiver object.

Field 1: Receiver Address

- Go to Fields & Relationships → click New



- Choose Data Type: Geolocation → click Next.
  Field Label: Receiver's Address
  Click Next → Next → Save & New.



- 3. Follow step 1 and 2 to create the remaining fields (Receiver Contact, Receiver Email, Sender, Parcel) of the Receiver object mentioned in the table below.

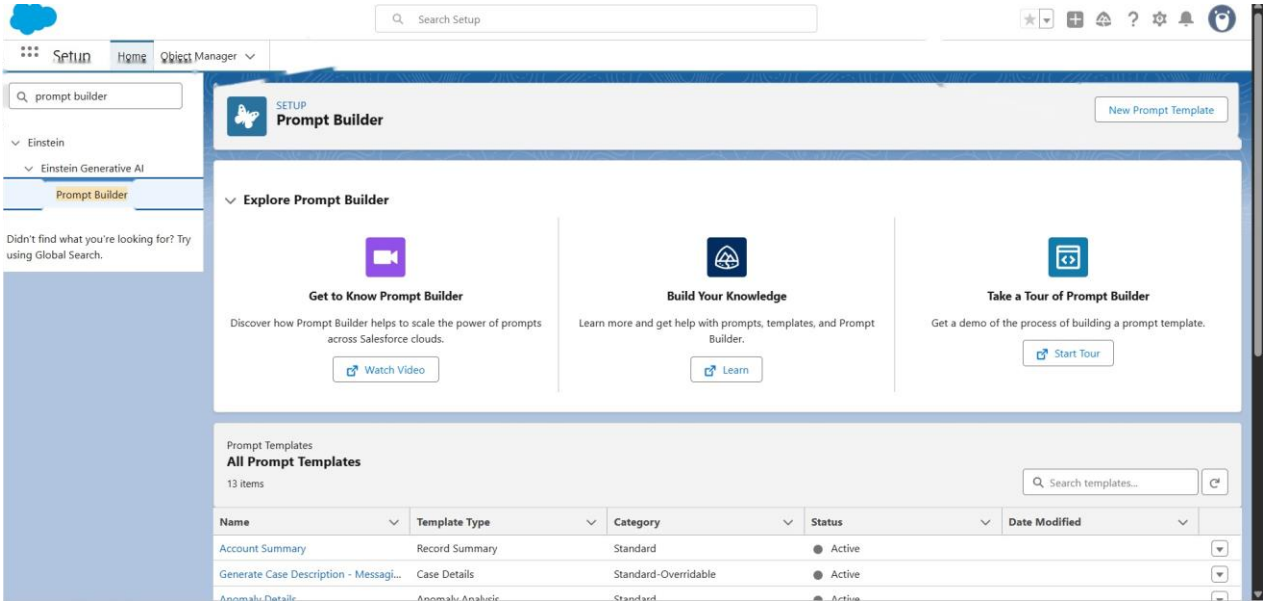
Field Name	API Name	Data Type	Description
Receiver Address	Receiver_Adress__c	Geolocation	Location in terms of longitude and latitude.
Receiver Contact	Receiver_Contact__c	Phone	To contact the Receiver
Receiver Email	Receiver_Email__c	Email	To send email Notification.
Sender	Sender__c	Lookup Relationship	Gets Details from Sender
Parcel	Parcel__c	Lookup Relationship	Gets Details from Parcel

Milestone 5 - Prompt Builder

Designing Prompts for Agent Building.

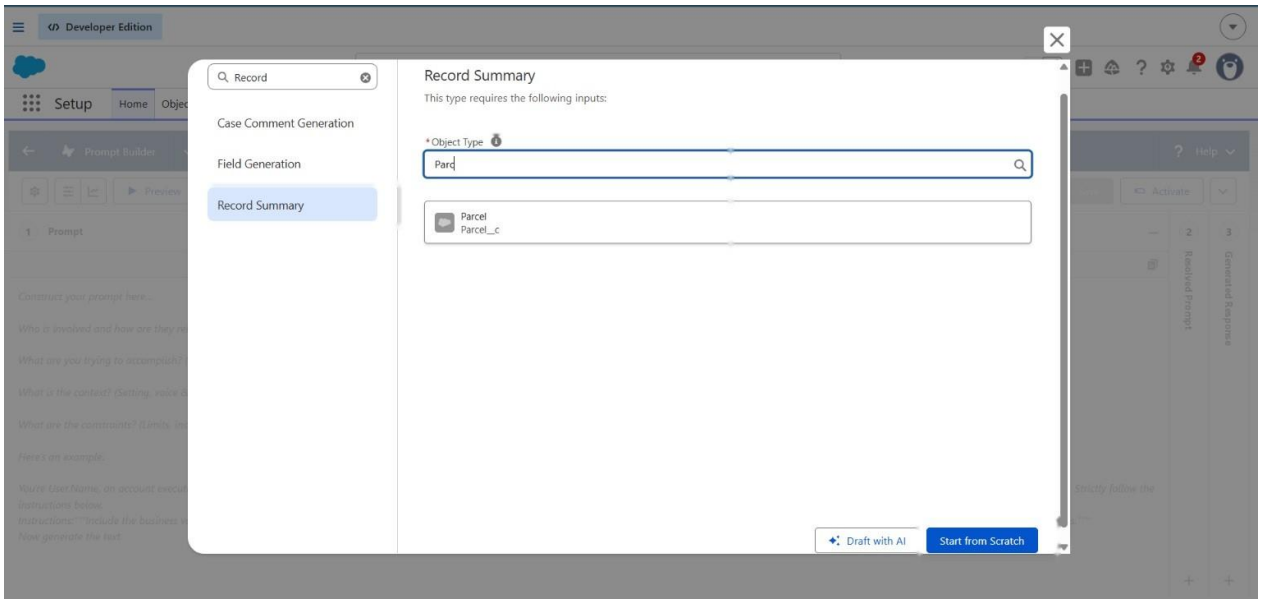
In Salesforce, go to Setup.

1. In the Quick Find box, type Prompt Templates (or in Agent Console, go to Agent Builder → Prompt Templates).
2. Click New Prompt Template.-- Record summary template



4. Enter the details

- Label: Retrieve Parcel Details
- API Name: Retrieve_Parcel_Details
- Description: Retrieves parcel details where Parcel_ID matches the user input.



5. Enter the Prompt Template Content

In the Prompt Template Text area, paste the following exactly:
Retrieve details from Parcel__c where Parcel_ID__c matches user input. Return response in this format:

Parcel Tracking Update

☐ Parcel Tracking Update

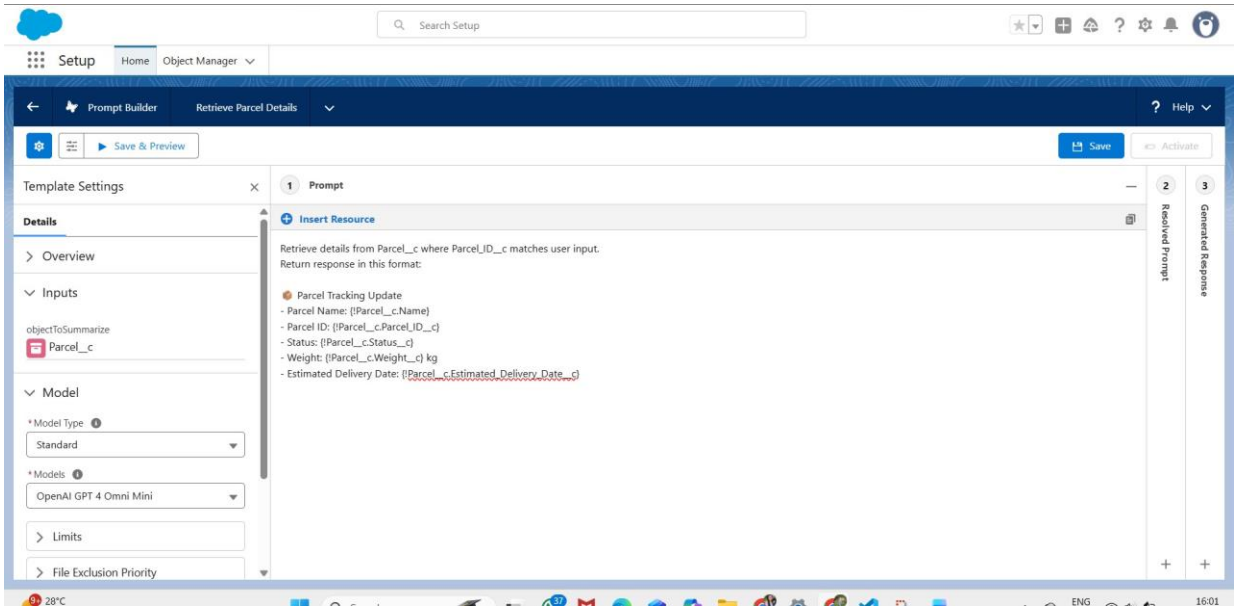
- Parcel Name:{!$Input:Parcel__c.Name}

- Parcel ID:{!$Input:Parcel__c.Parcel_ID__c}

- Status:{!$Input:Parcel__c.Status__c}

- Weight:{!$Input:Parcel__c.Weight__c}

- Estimated Delivery Date:{!$Input:Parcel__c.Estimated_Delivery_Date__c}



6. Add Resource Mapping
7. Scroll down to the Resources / Insert Resources section.

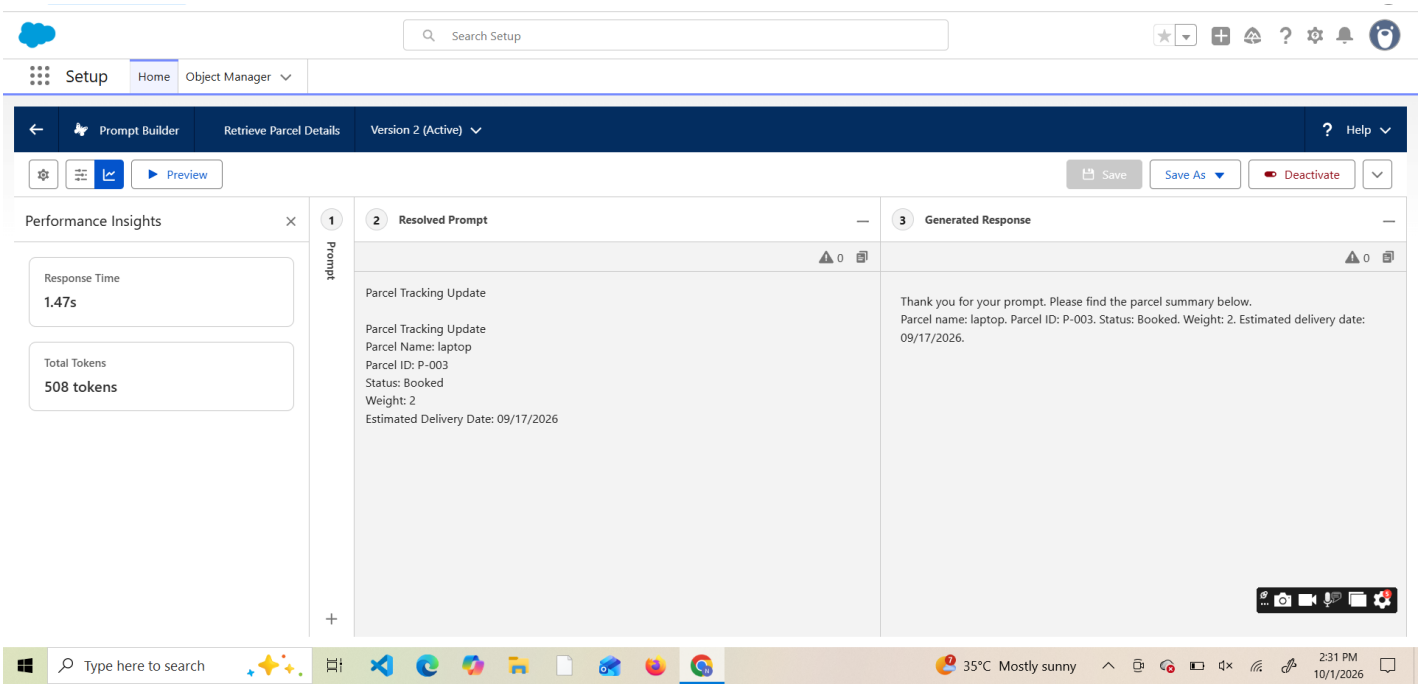
- Click Insert Resource → choose Object.
- Select Parcel__c.
- Add the following fields:
- Name
- Parcel_ID__c
- Status__c
- Weight__c
- Estimated_Delivery_Date__c

8. Click Insert Resource (or Add) for each one.
9. Click Save.
10. Then click Activate to make this Prompt Template available for your agent or flow.

Testing the Prompt

Test the Prompt:

1. Open your Agent preview.
2. Type: “Track parcel P-12345”.
3. It should return details from the Parcel record.

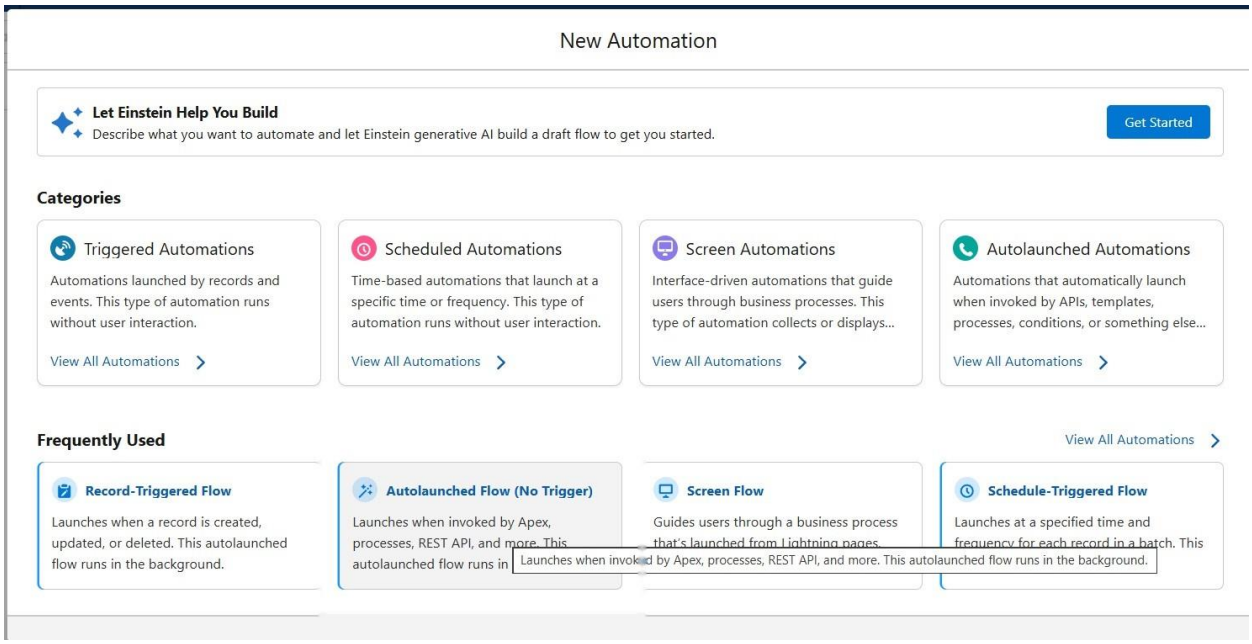


Milestone 6 - Flows

Create flows for capturing the response as shown.

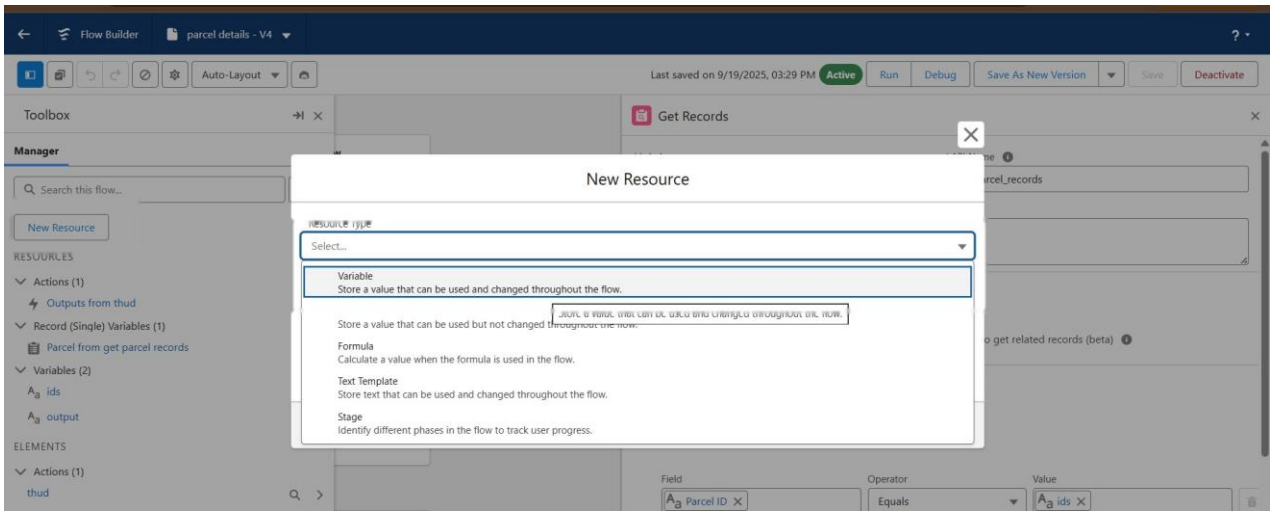
Go to Setup → search Flows → click New Flow.

1. Choose Auto-Launched Flow (No Trigger) → click Create

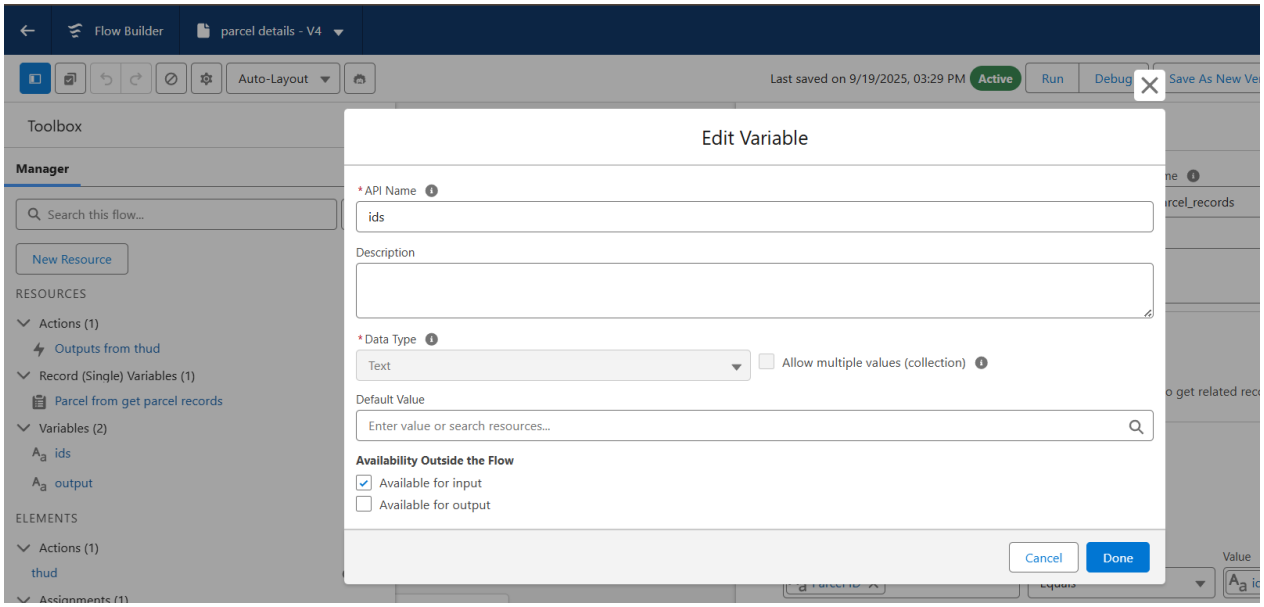


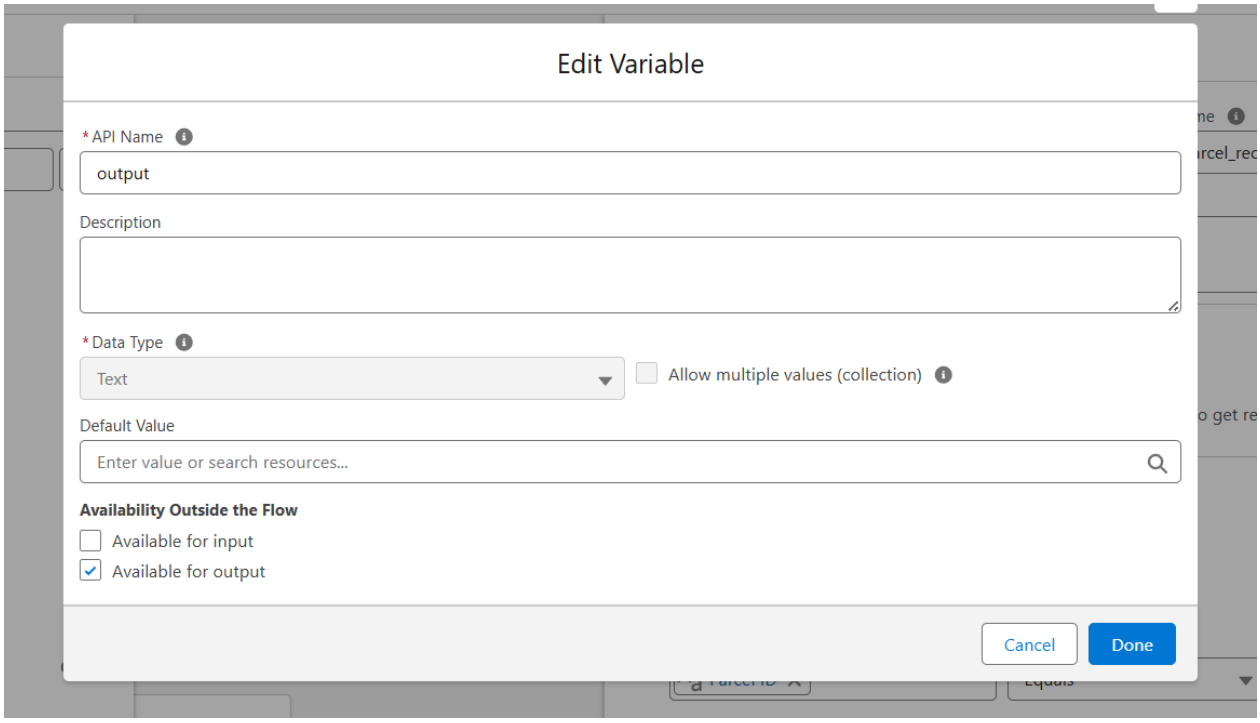
3. Add Input Variables (to allow Agent to pass values):

- Click On New Recourse →Variable → Available for Input.

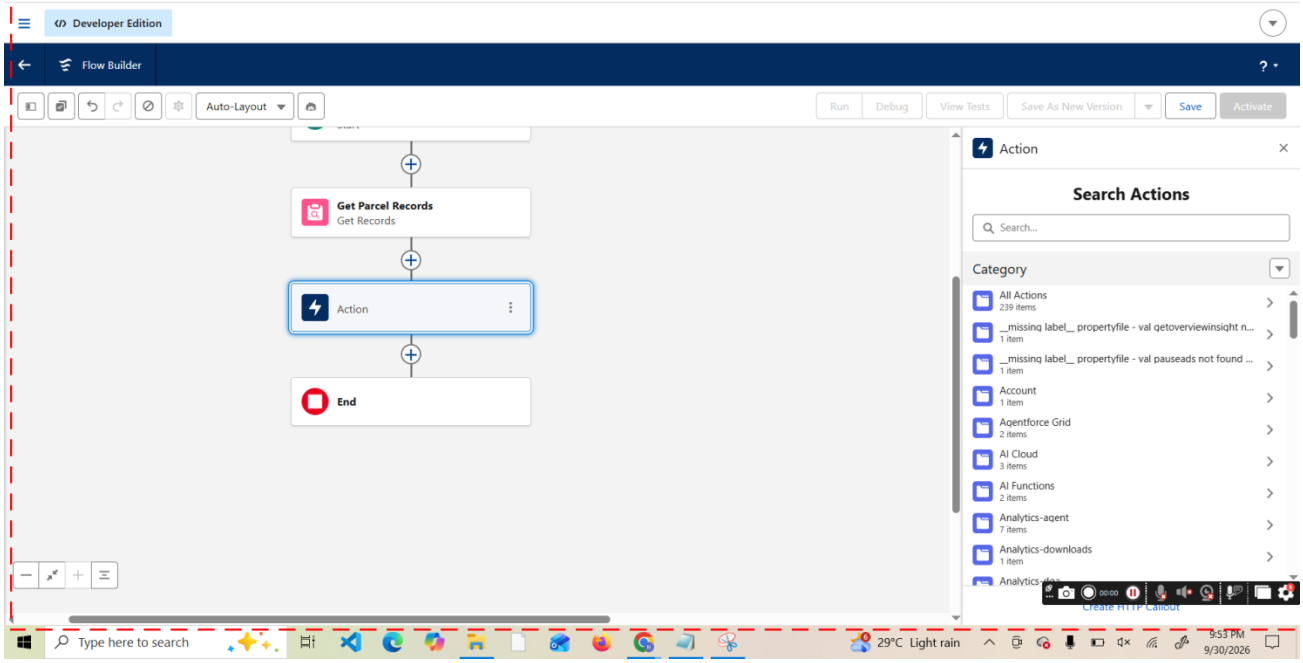


- Ids → Data Type: Text → Available for Input.
- Output → Data Type: Text → Available for Output.





4. Click on the Add icon to add Element: Get Records→ Label: Get Parcel Records

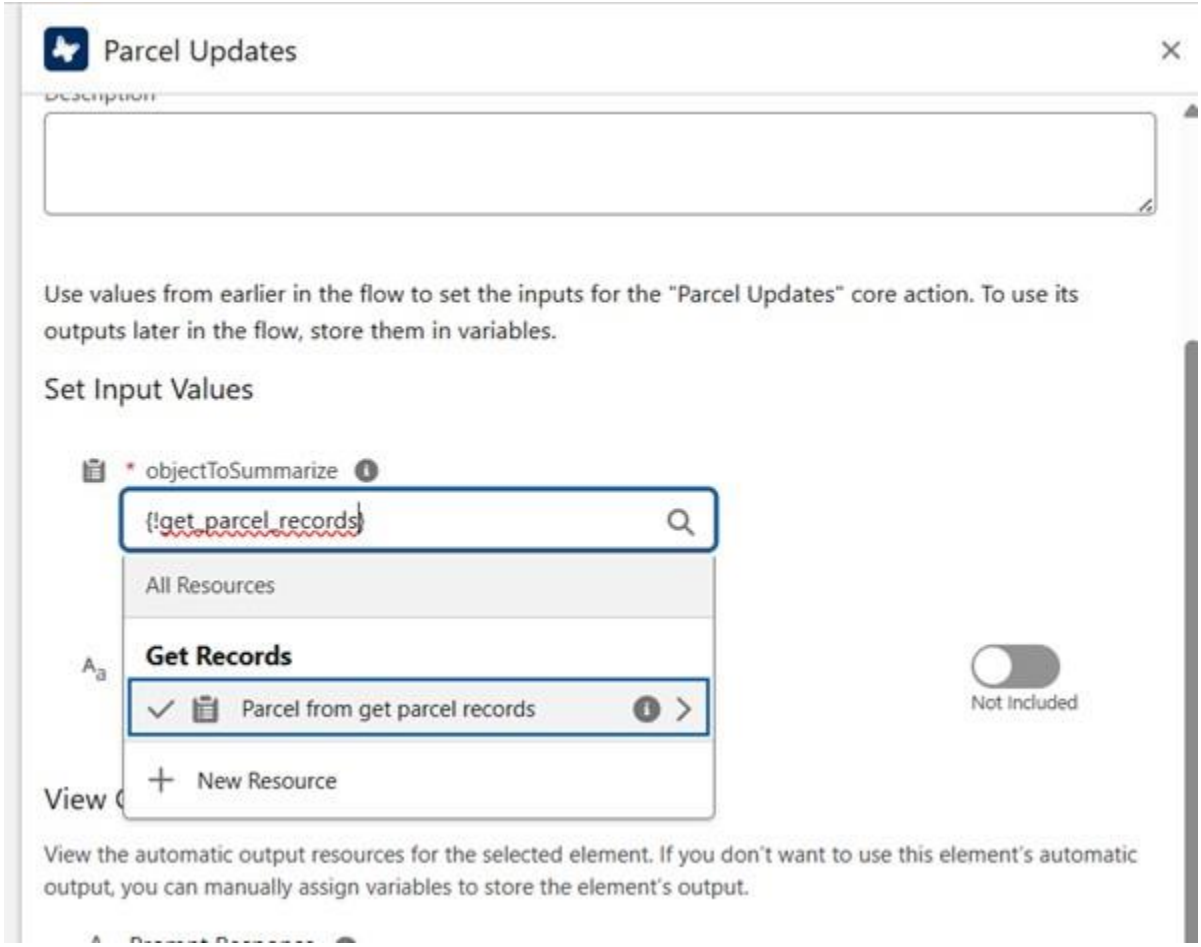


- Object: Parcel
- Field = Parcel_ID__c
- Operator = Equals
- Value = ids

5. Element : Action

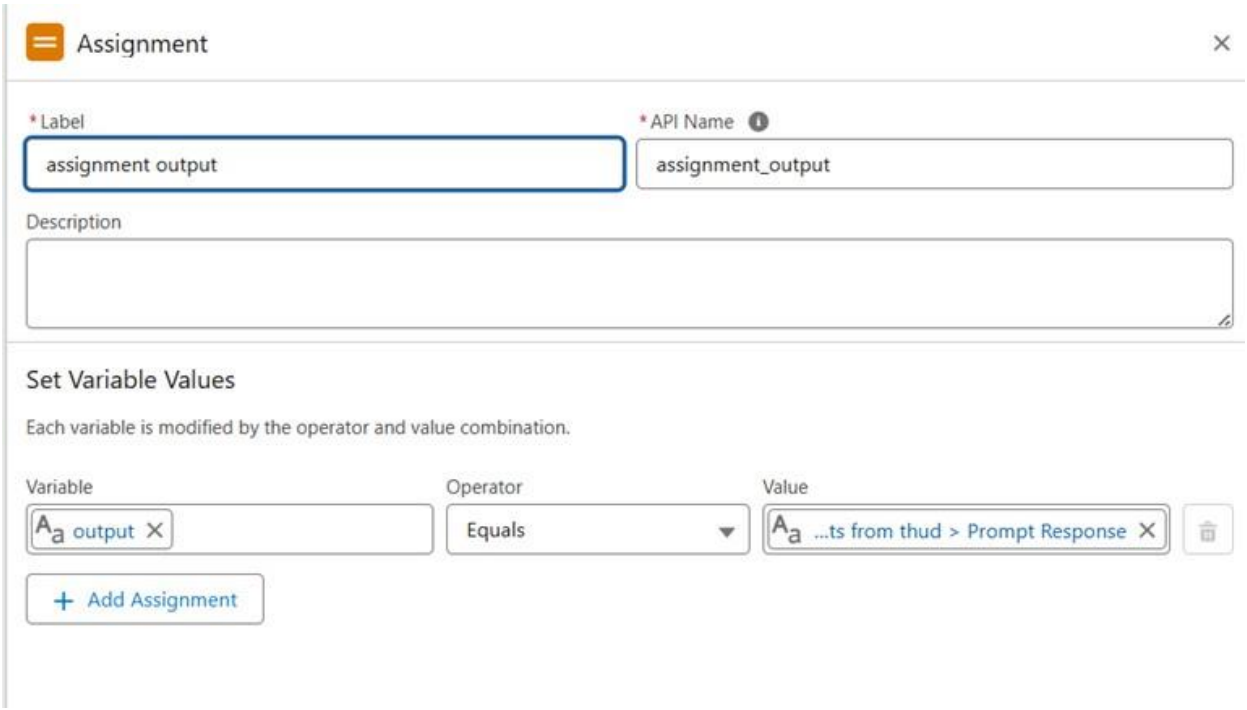
1. Find Retrieve parcel details
2. Click Retrieve parcel details

- Label: Parcel Updates.
- Object To Summarize : Parcel from get Parcel records.



6. Element : Assignment

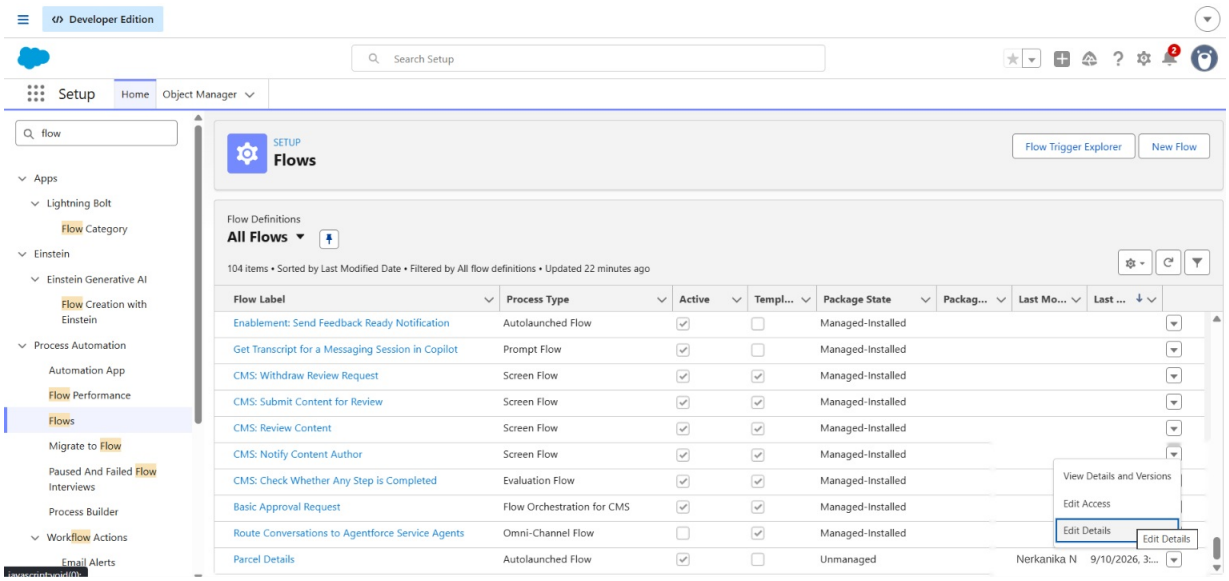
- Label: Assignment Outputs.
- Variable= Output
- Operator = Equals
- Value = Outputs from Action>Prompt Response



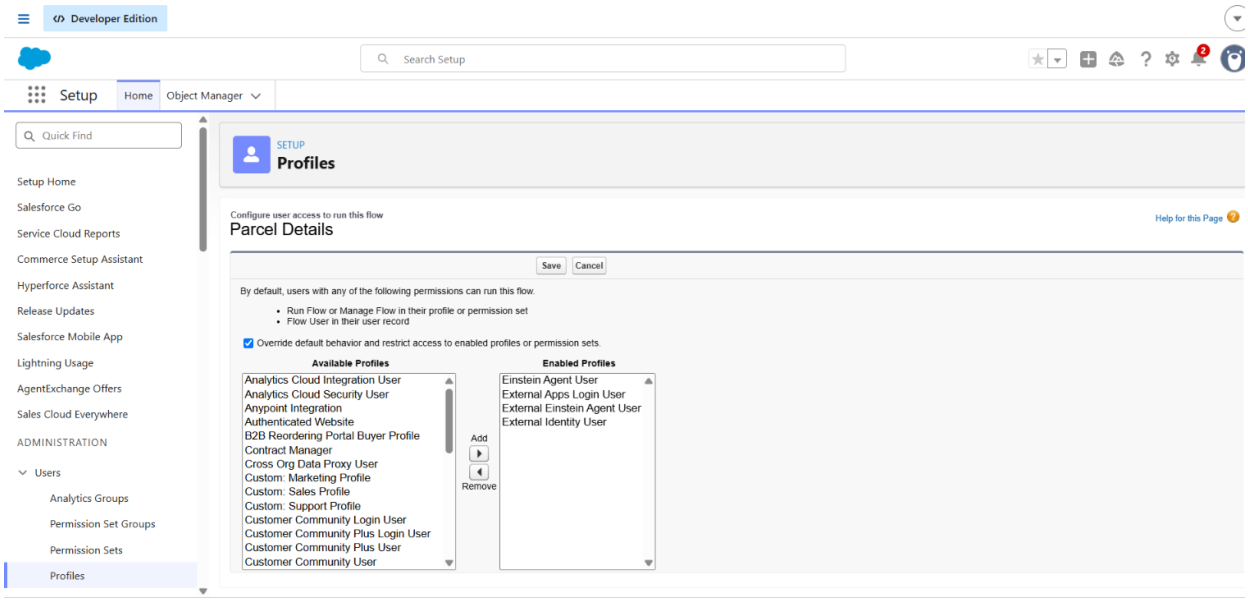
7. Connect Start → Get Records → Action → Assignment → End.

8. Save Flow → Name: Parcel Details.

9. Activate Flow.
10. Once the flow is saved go back to the flows, Click on the drop down near the flow that is just created, click on edit access.



11. On the top look for flow user access and then assign these available profiles, and click save.



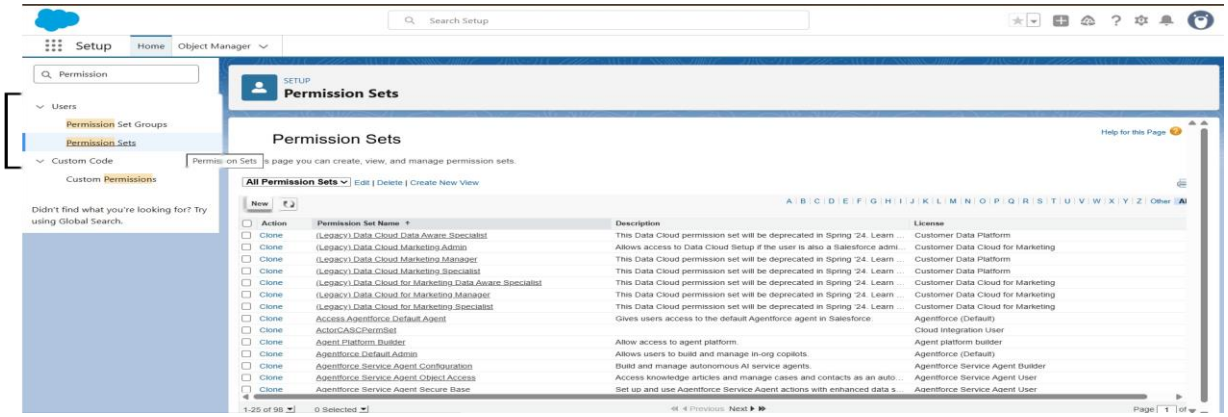
Phase 3: UI/UX Development & Customization

To enable agents to efficiently manage parcel operations and ensure secure access control using permission sets and AI-driven agent experiences built with AgentForce.

Milestone - 1 Creating Permission Sets

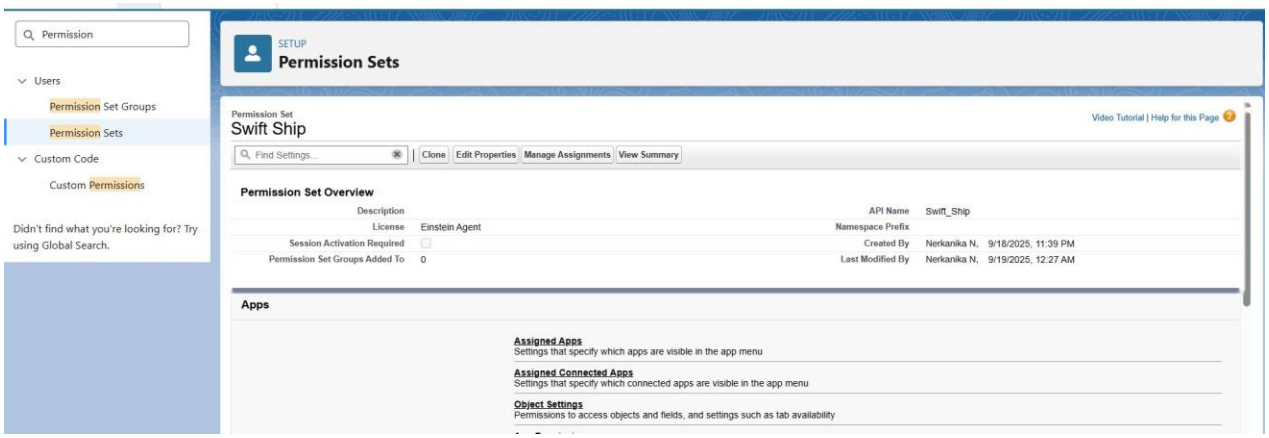
In the Quick Find box (left side), type Permission Sets.

1. Click Permission Sets under Users.

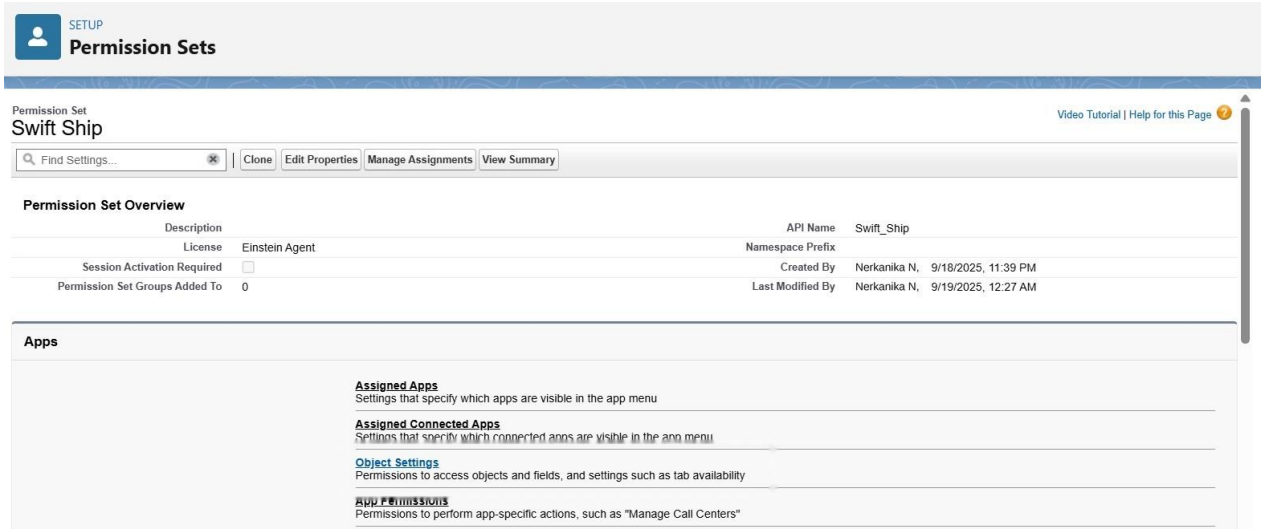


1. Click the New button.

2. Enter the following details:
    - Label: Swift Ship
    - API Name: auto-fills
    - License: Choose Einstein Agent.
3. Click Save.



1. Inside the Permission Set page → under Apps → Object Settings, click Object Settings.



1. Search and click each object one by one
2. Click Edit.

Enable the following:

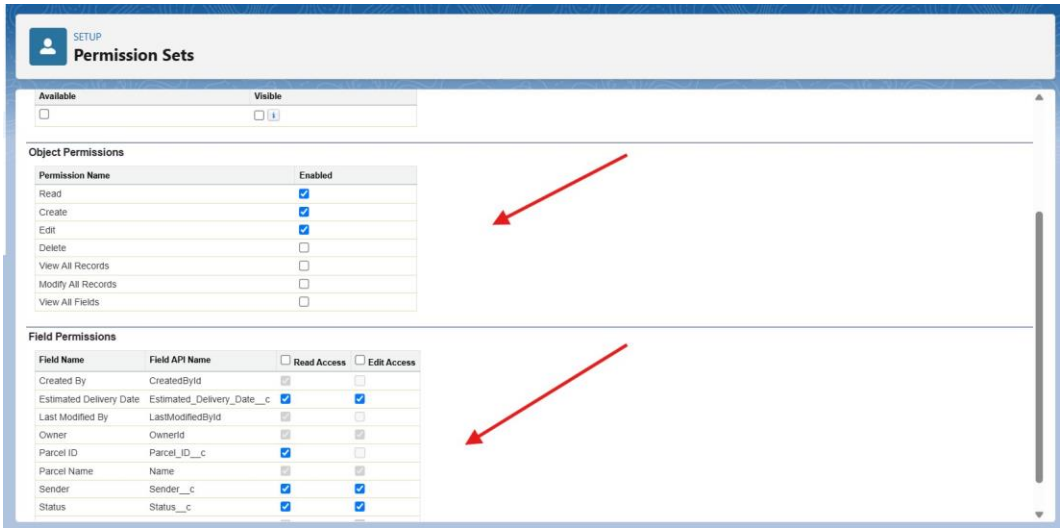
✅ Read
✅ Create
✅ Edit

(Optional: Delete, if you want agents to remove records)

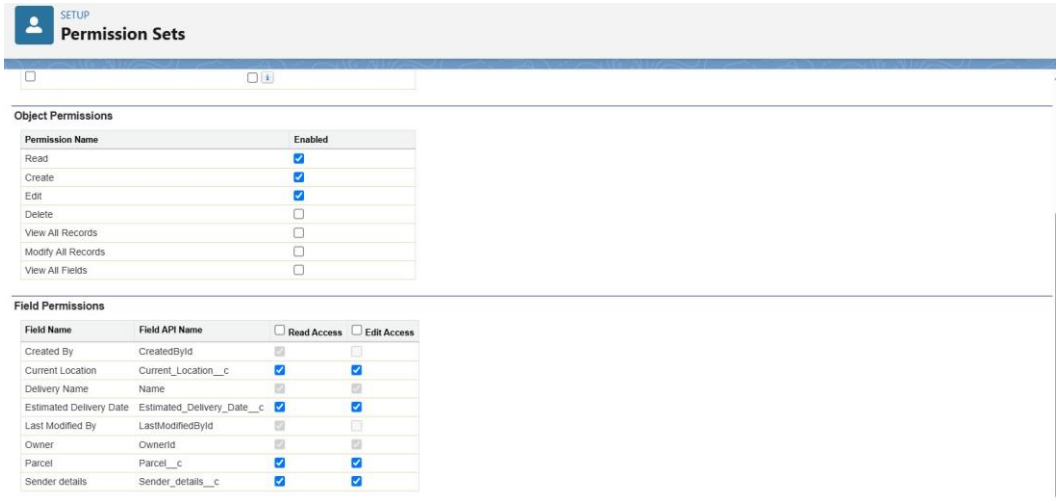
Under Field Permissions, ensure at least Visible is checked for all key fields.

Click Save after each object.

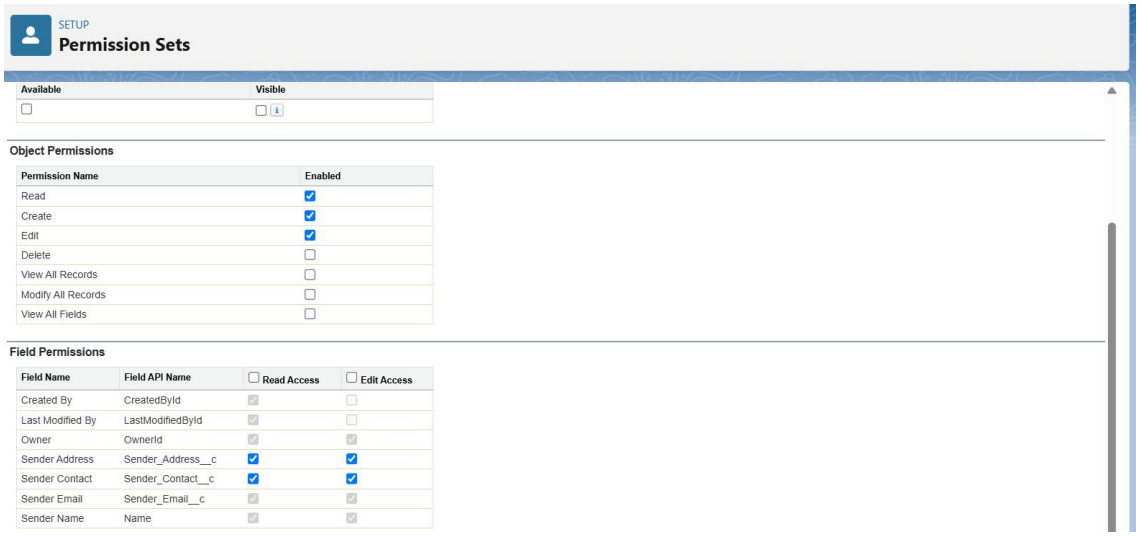
Parcel__c



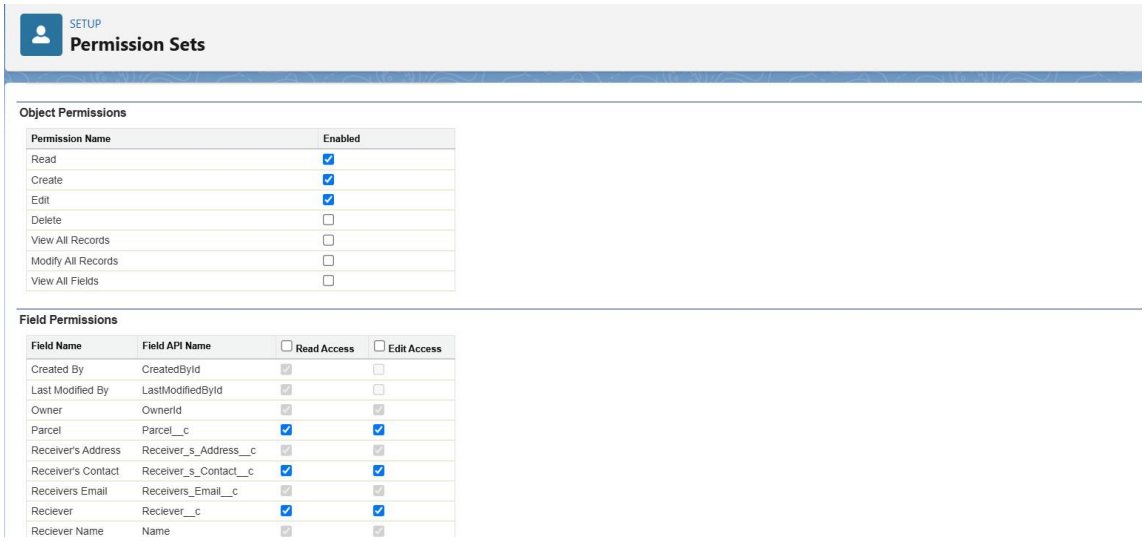
Delivery__c



Sender__c



Receiver__c



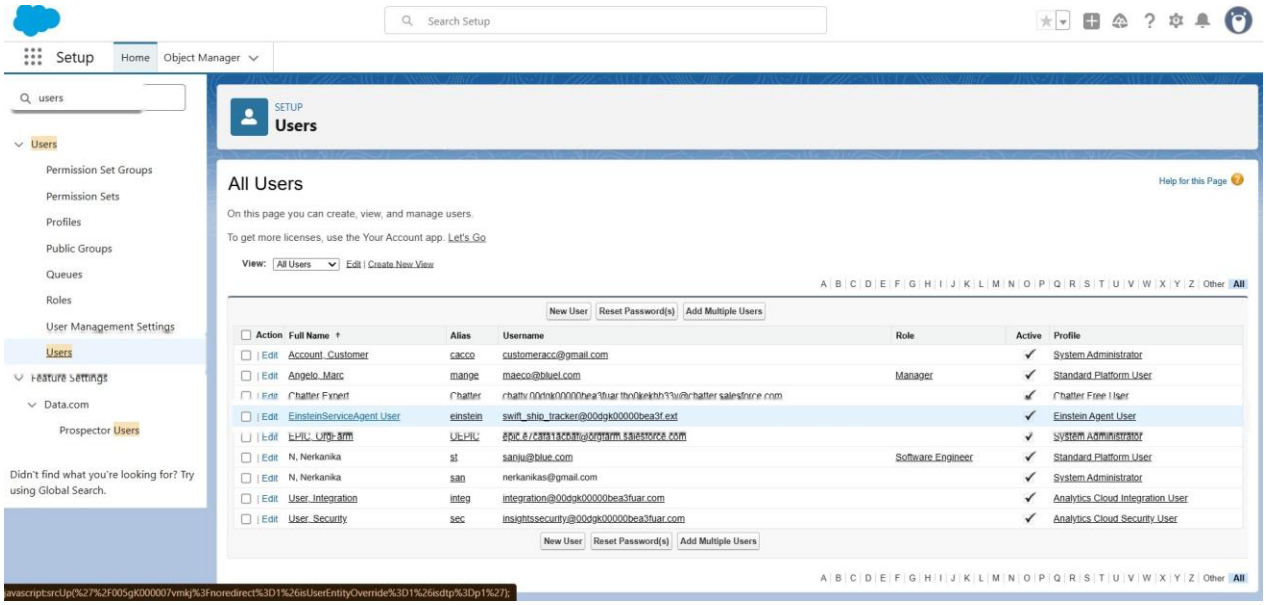
Milestone - 2 Assigning Permissions to your Agent user

In the Quick Find box on the left, type Users.

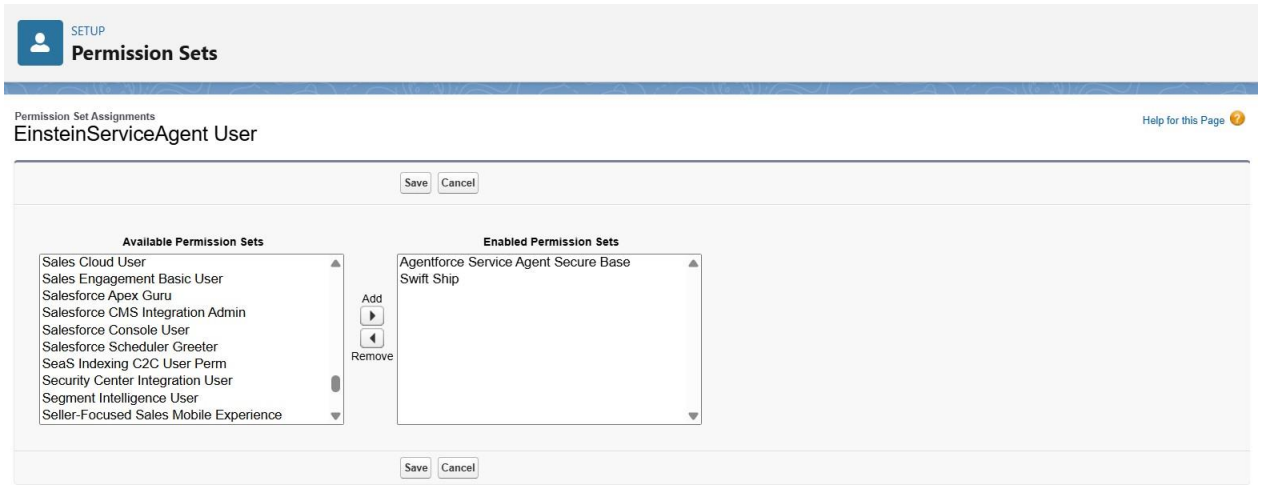
1. Click Users under the Users section.
2. Open the Agent User Record

From the list of users, find the Einstein Service Agent user to represent your Agent (e.g., “SwiftShip Agent”).

Click on the Full Name of that user to open the record.



4. Scroll down to the section named Permission Set Assignments.
5. Click the Edit Assignments button.

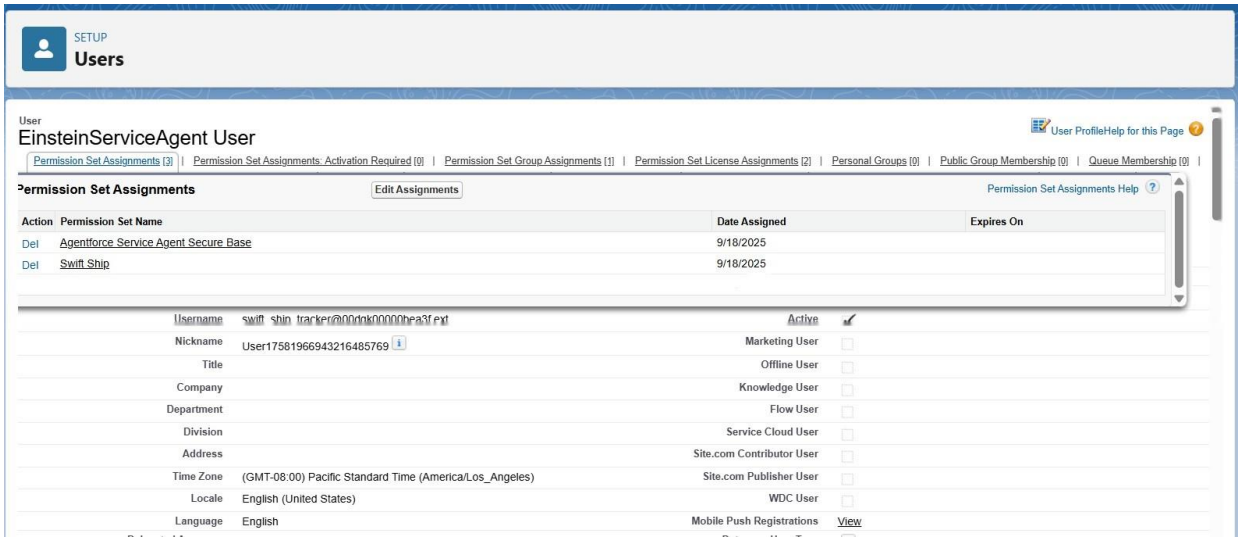


1. On the left side (Available Permission Sets), find and select your permission set:

- SwiftShip

Click the Add → button to move it to the right-side list (Enabled Permission Sets).

Click Save.



Milestone -3 Org Setup for AgentForce

Enable salesforce Default agent from Salesforce Go

1. From Quick find box→ search “Salesforce Go” → Click on it
2. Search for Default agent —> click get started→ and turn on
3. And refresh the page

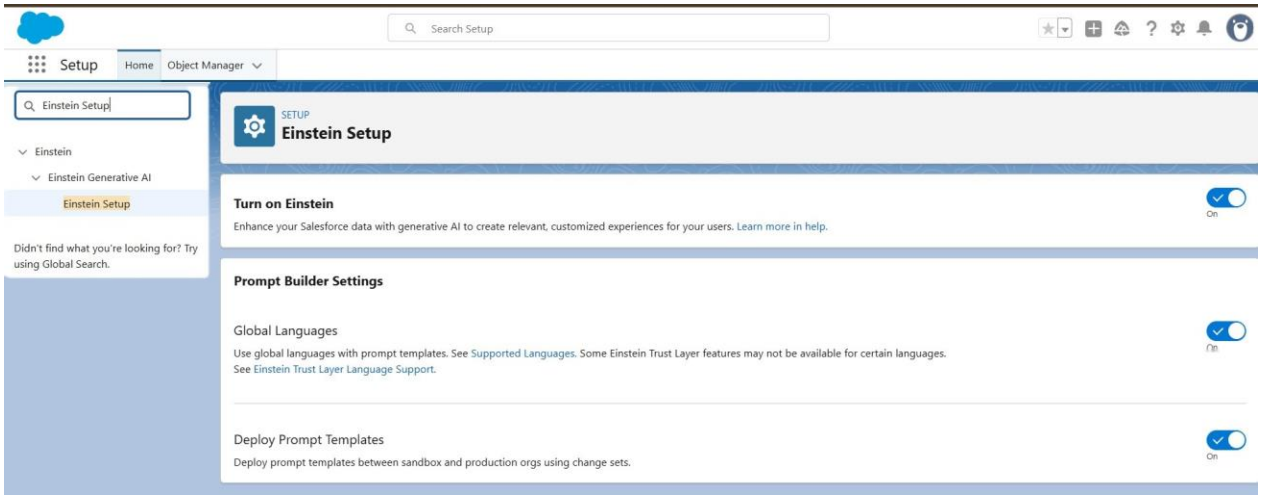
Enable Einstein Setup

Go to Setup → Quick Find → Einstein Setup.

1. Open Einstein Setup → click Get Started if prompted.

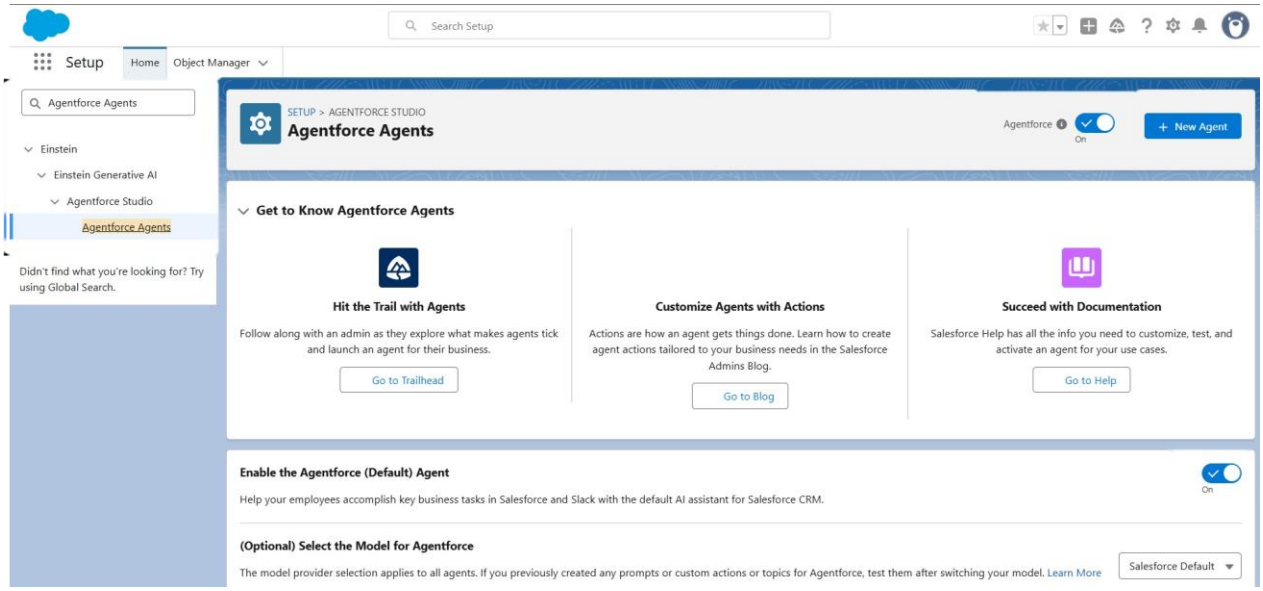
Toggle On these key services:

- Turn On Einstein
- Global Languages
- Deploy Prompt Template



3. Refresh your browser page.
4. Go to Setup → Quick Find → AgentForce Agents.
Toggle On these key services:

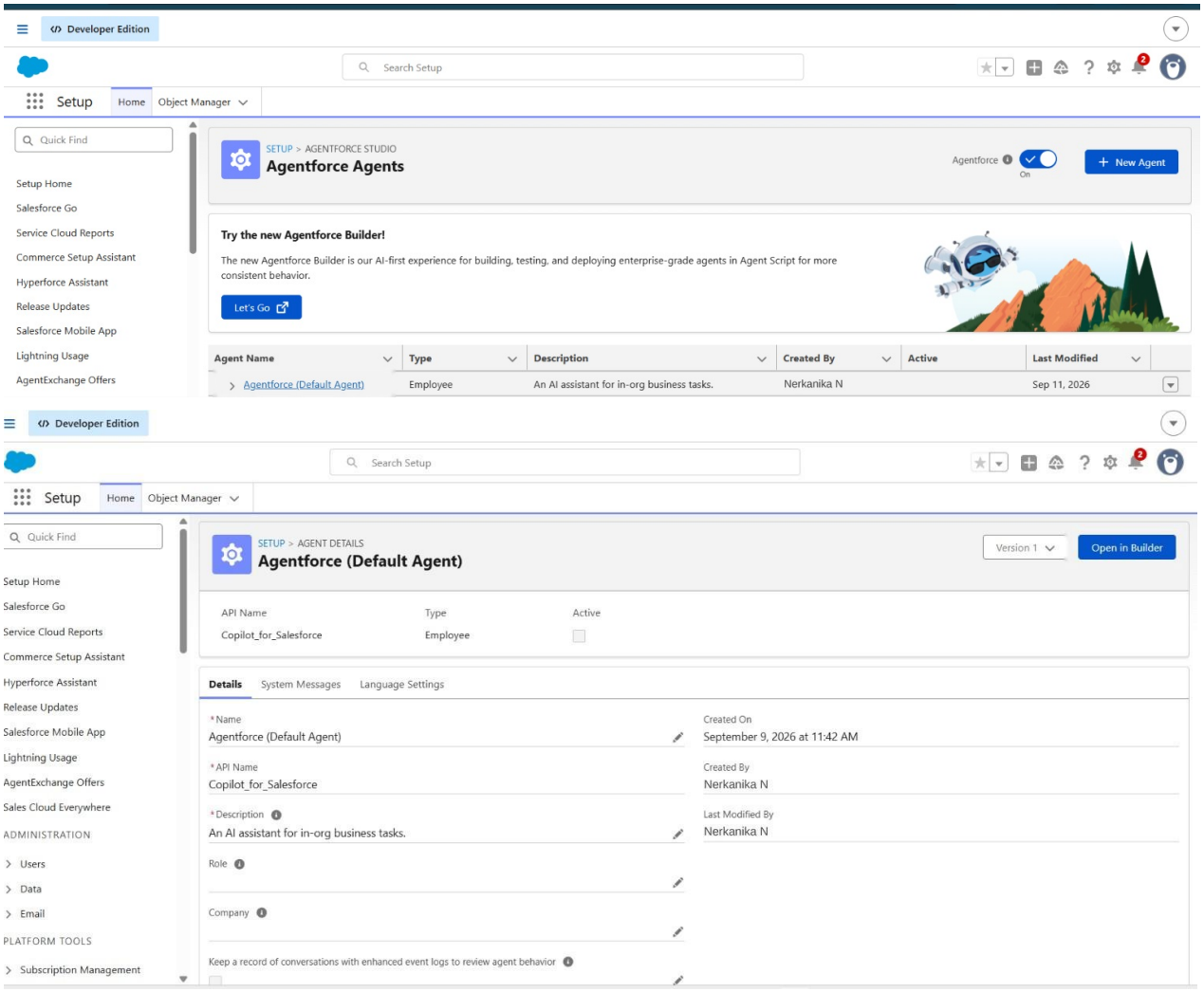
- Agentforce
- Enable Agentforce Default Agent



Milestone - 4 Building The Agent

1. In Setup, search for Agentforce Agents in Quick Find.

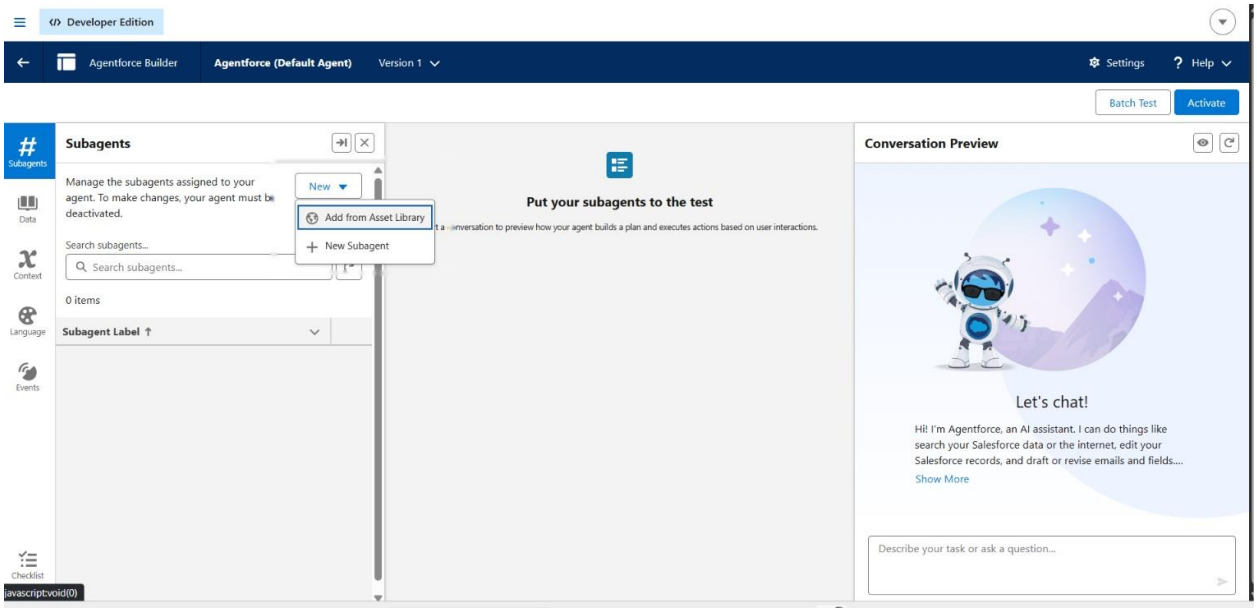
2. Click Agentforce Agents, then click on the default Agent, Then click on Open in Builder.



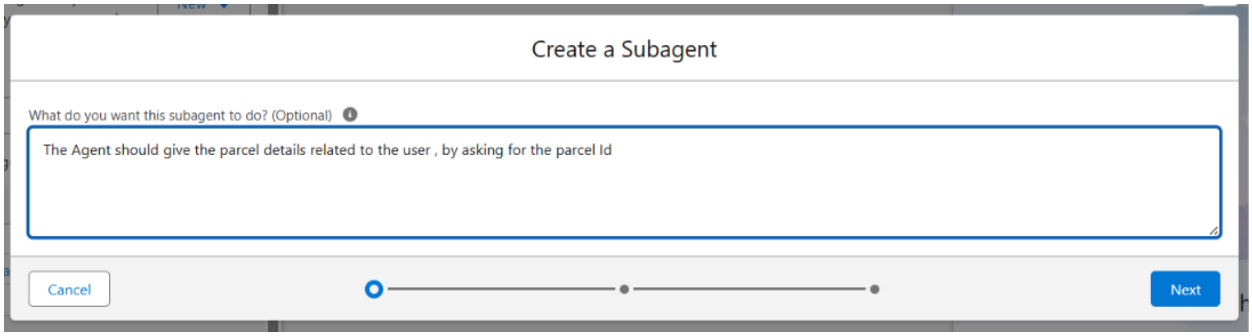
Assign Sub Agents to the Agent

On the subagents Click on New.

1. Click New Subagent

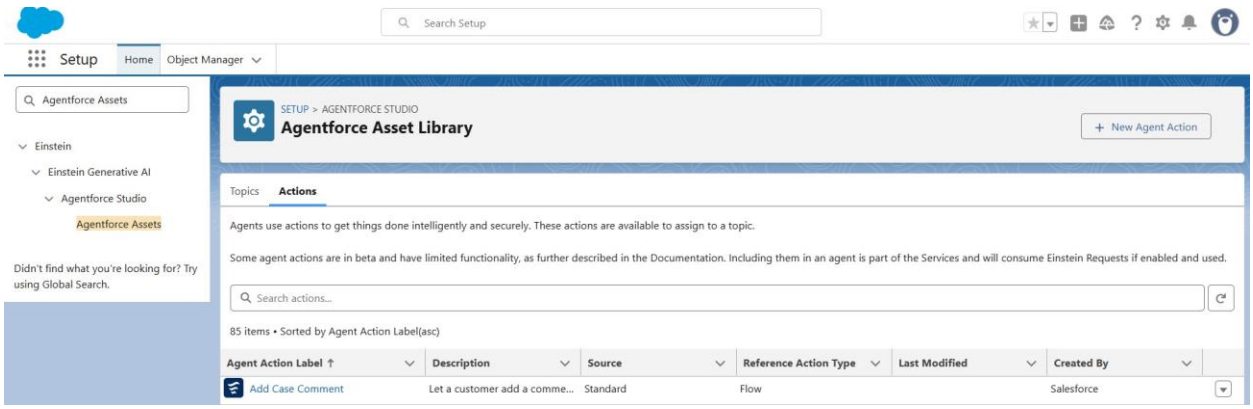


3. Create a new subagent.
4. Fill in the details as shown.

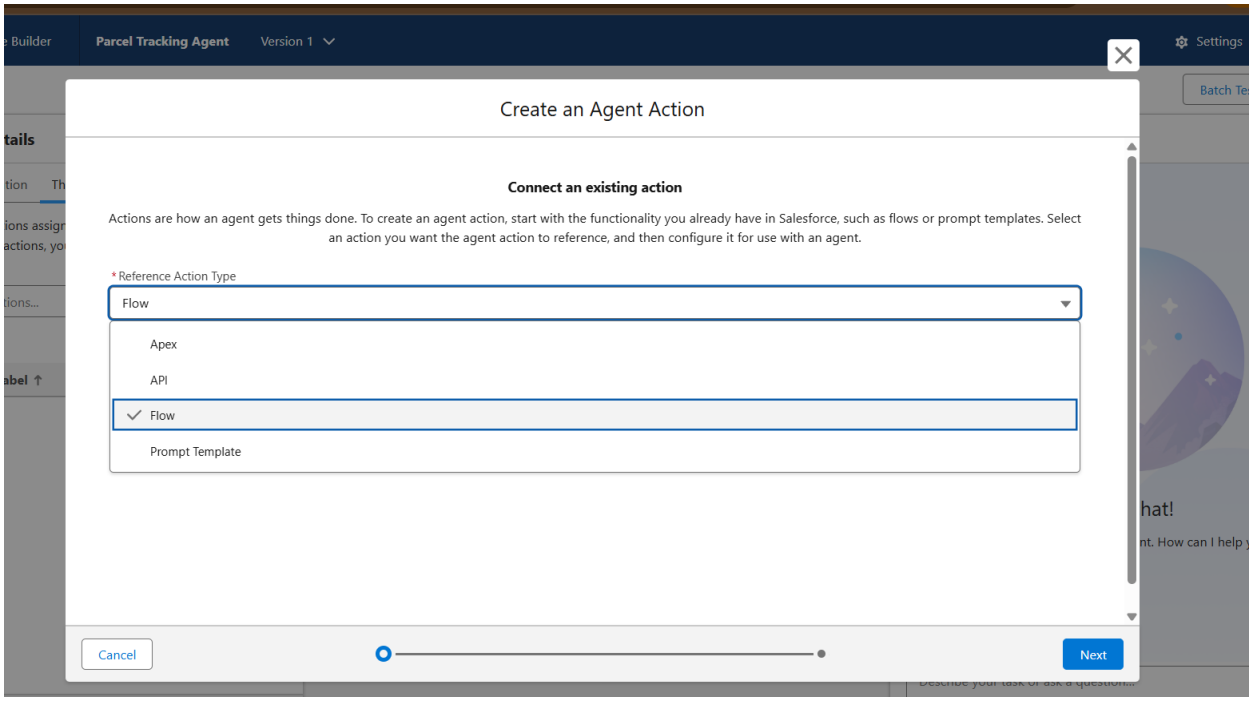


Adding Actions to Agentforce Assets

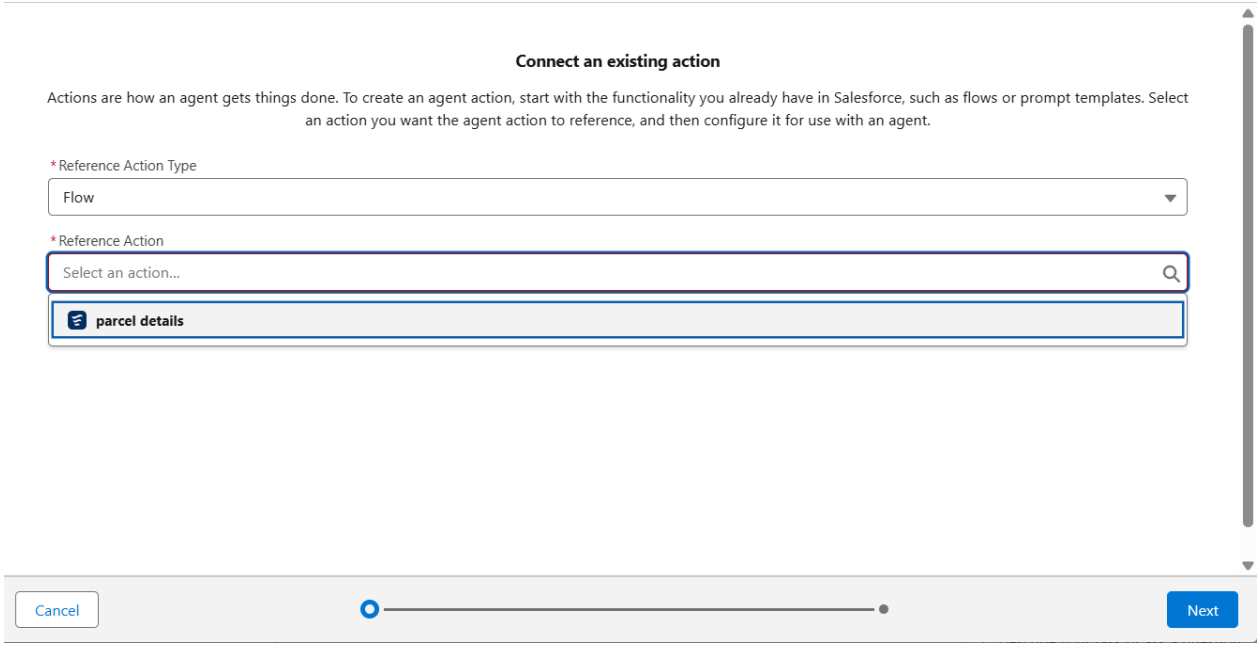
While still on the Agent Builder in the topics, Click on the Topics Actions



2. Click on New Agent Action.
3. For Action type, Click on Flow then Next.



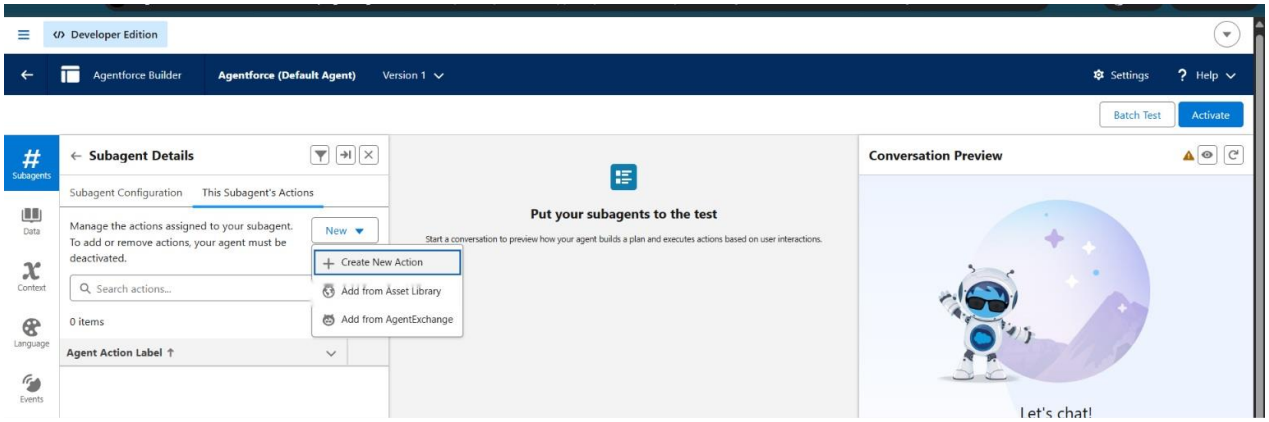
4. For Reference Flow use the recently created flow, click Next.



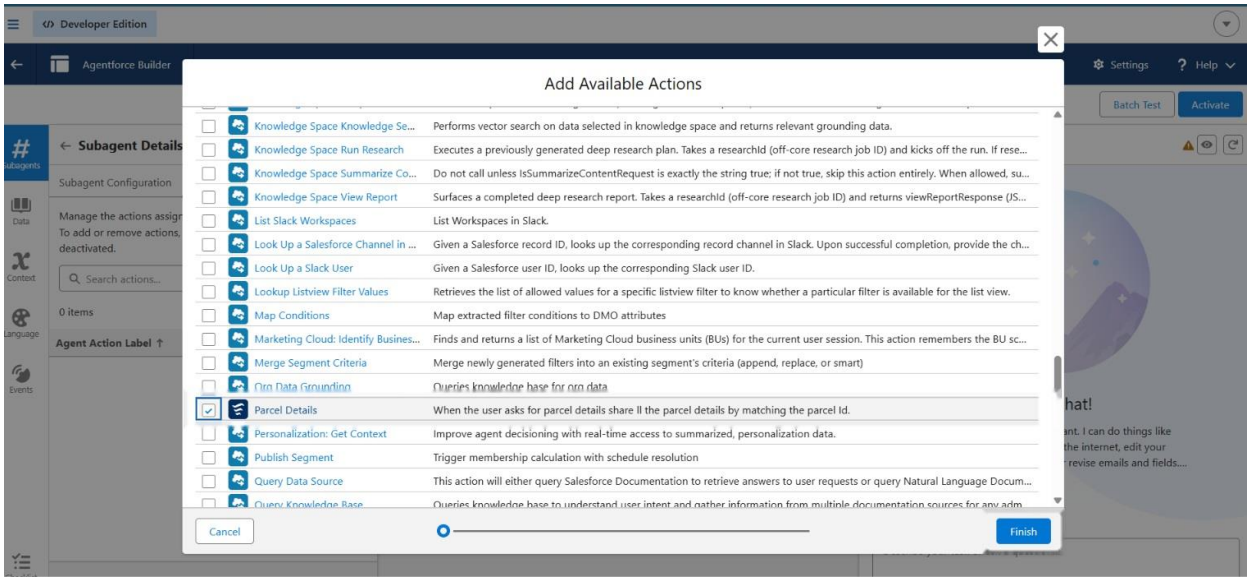
5. Fill the details for Agent Configurations.

Assign Subagents Action

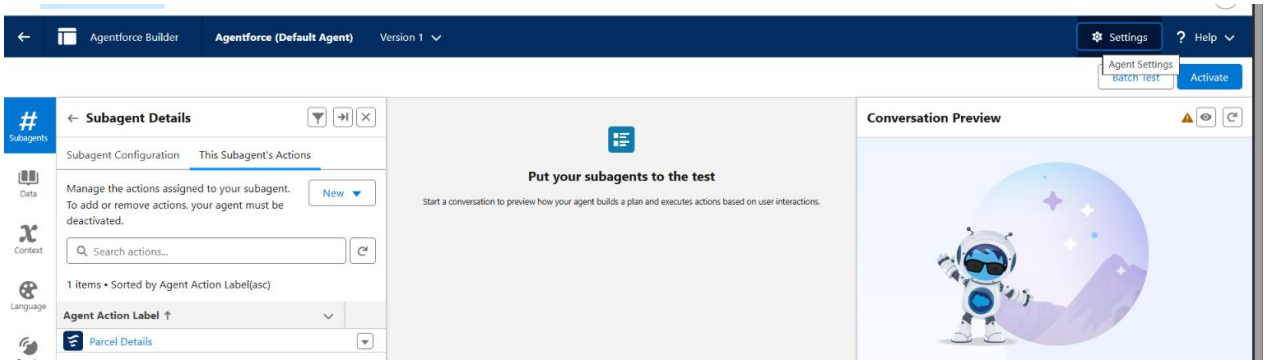
While on the Agent Builder in Subagent, Click on the Parcel Updates Subagent



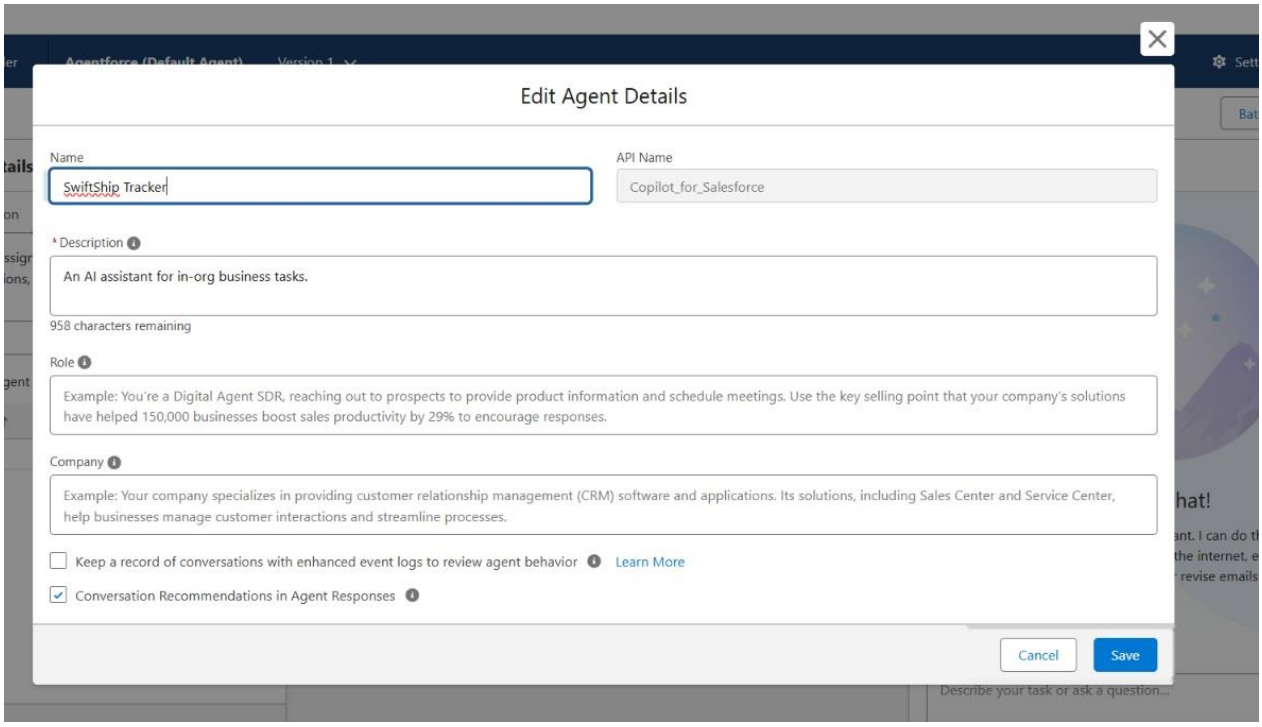
2. **Click on the subagents Actions, then select Add from asset library.**



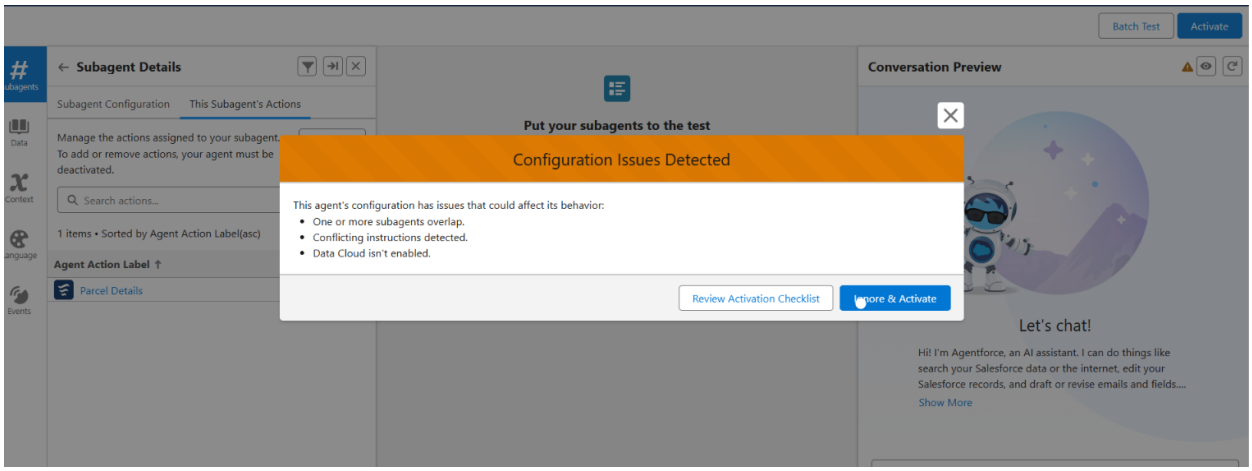
3. **Select the flow action that we created and click on finish.**



4. **Click on settings , and rename your agent to** SwiftShipTracker



5. **Click Save.**



6. Click on Ignore and Activate, then Activate.

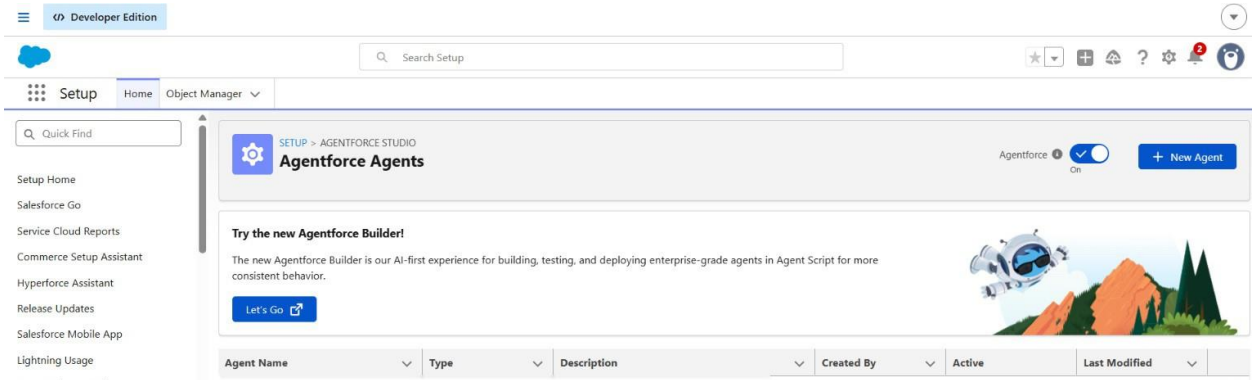
Phase 4: Data Migration, Testing & Security

To validate the functionality and reliability of the Parcel Tracking Agent created in Agent Builder. This phase ensures that the agent correctly recognizes topics, triggers the assigned Flow (SwiftShipTracker), retrieves or updates parcel data, and performs all intended actions without access or permission issues. The goal is to confirm that the AI-driven agent operates seamlessly within Salesforce and provides accurate, real-time responses to parcel management queries.

Verify Agent

Go to Setup → Agentforce Agents.

1. Open the Agents list and confirm that your agent Name: Parcel Tracking Agent, appears



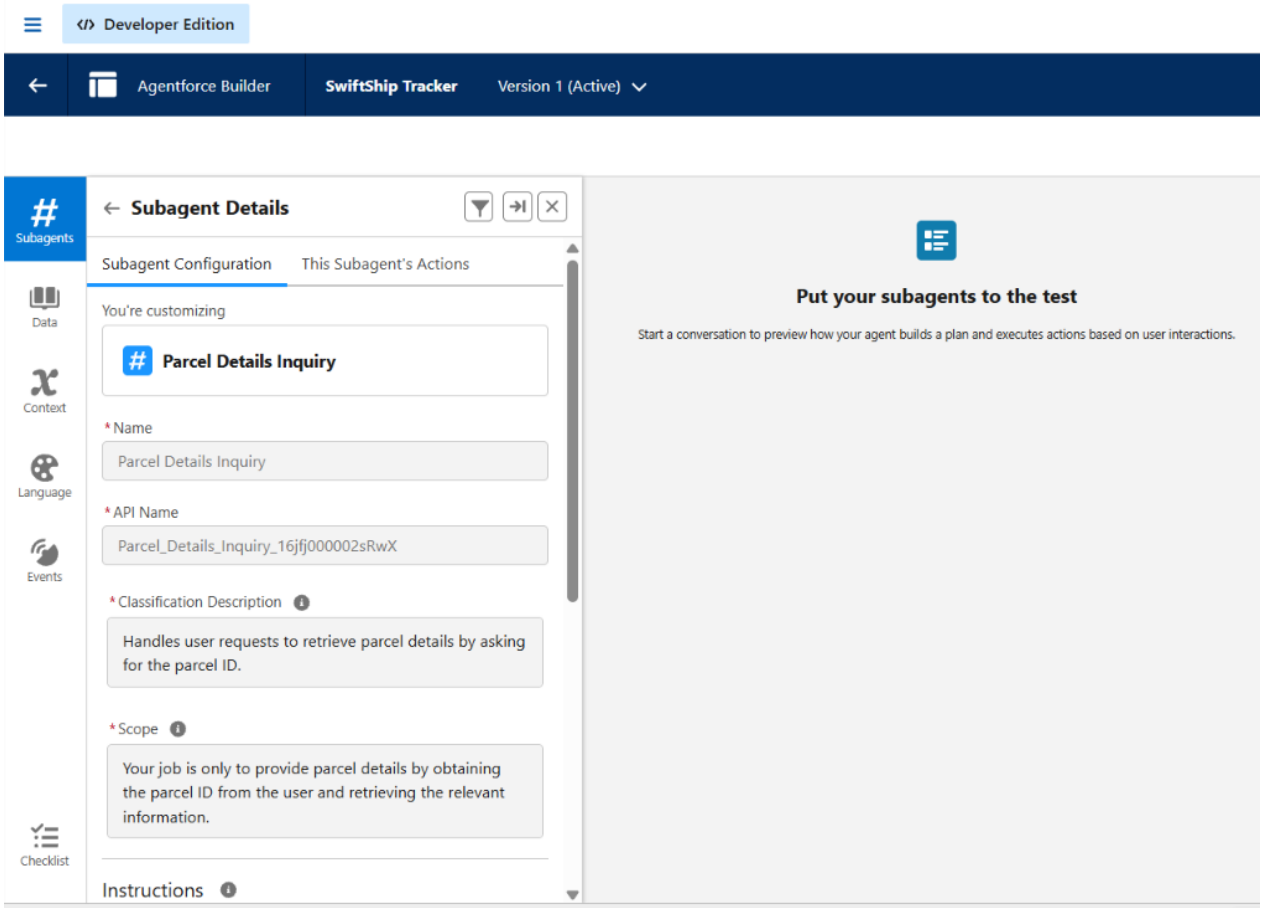
Validate Assigned Topic

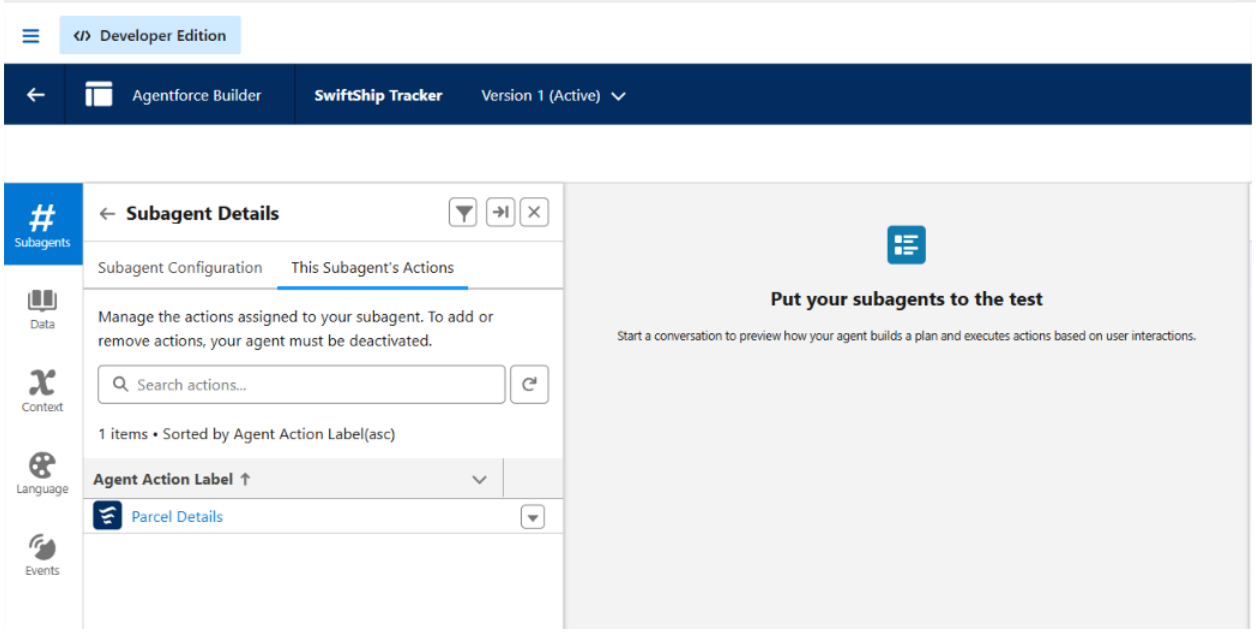
In Agent Builder → SubAgent, confirm that the SubAgent you created (e.g., Parcel Tracking or SwiftShipTracker) is listed.

Open the topic to verify:

- It contains a prompt or description related to parcel management.
- The Action/Flow (SwiftShipTracker) is linked to this topic.

2. Click Save and ensure the topic is Active.

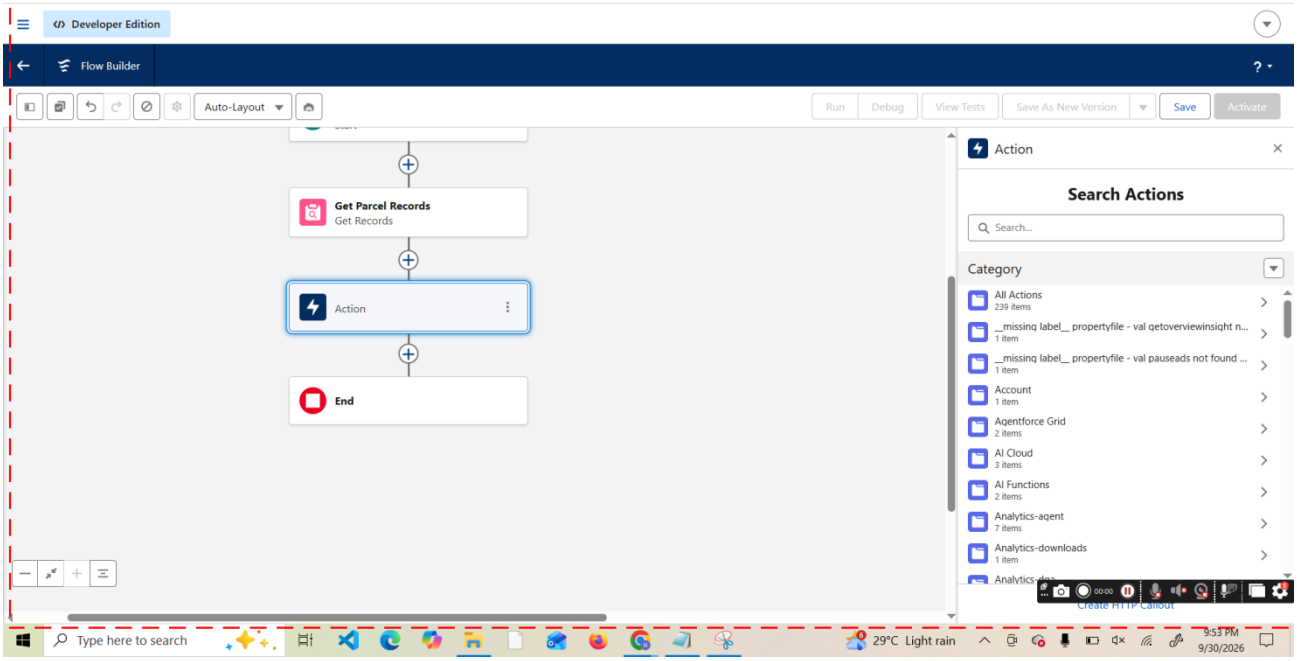




Check the Flow Association

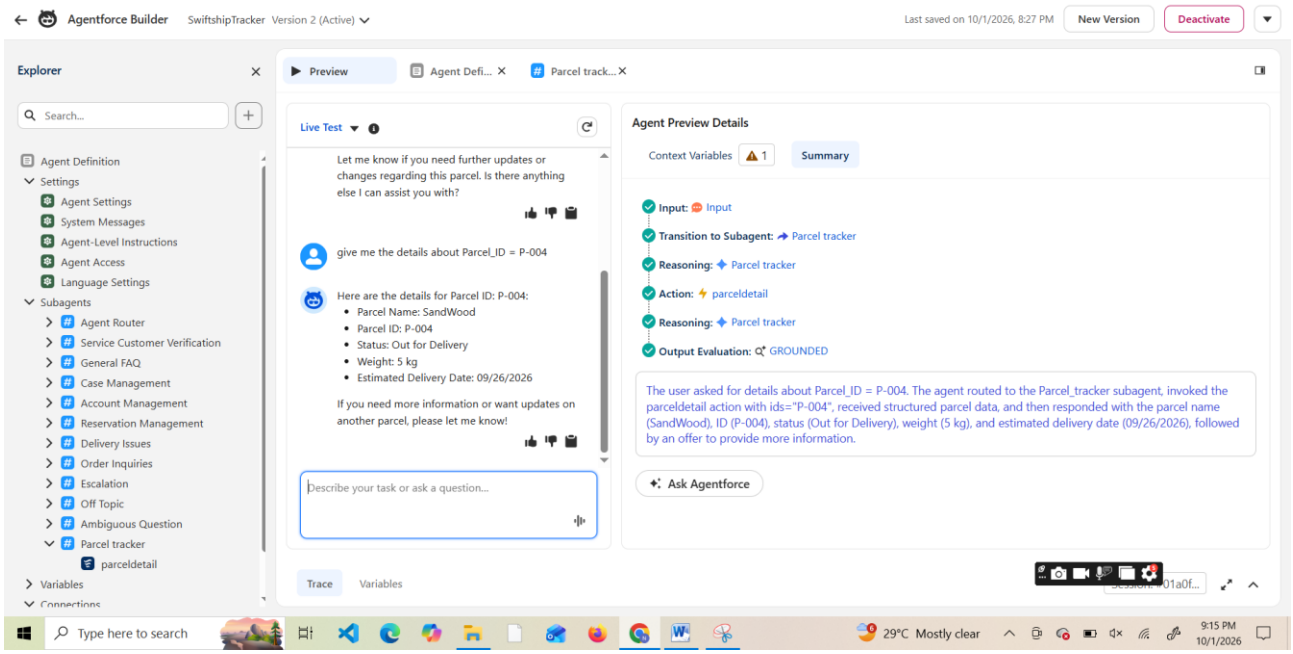
Go to Setup → Flows.

1. Locate the Flow named SwiftShipTracker
2. Confirm:
    - The Flow is Activated.
    - The Flow includes screens or actions that handle:
        - Parcel ID input
        - Parcel details retrieval
        - Status update or confirmation message
3. Open the Flow's Advanced Settings → ensure Run in System Context (Without Sharing) if data access is restricted.



Activate and Test the Agent

1. Return to Setup → Agent Builder → Parcel Tracking Agent.
2. Click Activate (if not already activated).
3. Once active, test using Preview / Try it Out mode:
   Type a query such as:
   "Track parcel ID P12345" or
   "Update delivery status for parcel P67890 to Delivered."
4. Observe the agent's response.



Validate Data in Salesforce

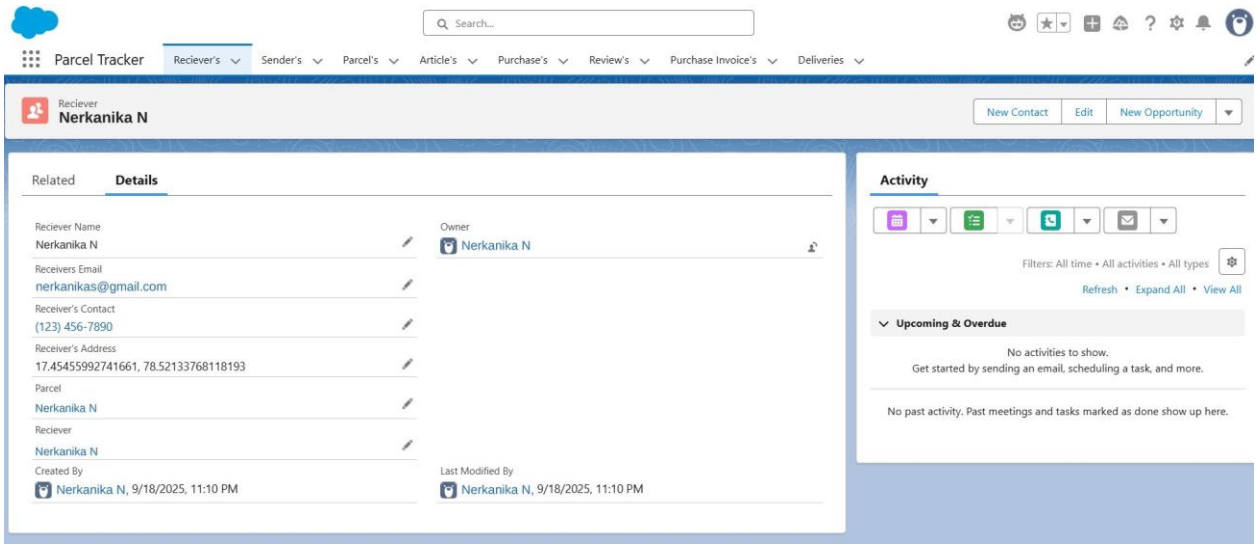
1. After the agent runs the Flow:

- Go to App Launcher → Parcels (Parcel__c).

Search for the Parcel ID you used during testing.

Verify:

- The Status or related field is updated.
- Any Flow-created records (bookings, updates) appear correctly.



Phase 5: Deployment, Documentation & Maintenance

- In the scope of our SwiftShipTracker (AgentForce) project implementation:

- The purpose of this phase is to validate that the Parcel Tracking Agent, created using Salesforce Agent Builder, works as intended within the Salesforce Developer Org environment.

- The main goal is to understand how an AI-powered Service Agent can be tested to ensure correct topic recognition, Flow execution, and data retrieval from Salesforce custom objects.

- Testing is conducted within the Developer Org to simulate how the Agent would perform in a live setup, ensuring that permissions, flows, and interactions function correctly before any deployment stage.

- Real-world production testing (UAT or sandbox deployment) is not performed in this project because:

    - Developer Edition orgs are standalone environments and do not support production-linked testing environments.
    - Enterprise-level AgentForce deployment involves sandbox testing, staging environments, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

Maintenance, Monitoring & Troubleshooting

In the SwiftShipTracker (AgentForce) project, ongoing maintenance ensures that Parcel, Customer, and Delivery data remain accurate, synchronized, and up to date. Regular monitoring helps detect issues such as misrouted agent queries, failed Flow executions, or performance delays in AgentForce responses. Troubleshooting activities include resolving configuration errors, updating Flow logic, refining agent topics, and improving prompts to enhance user interactions.

Although this project is implemented within a Salesforce Developer Org for learning purposes, in real-world enterprise deployments, these maintenance and monitoring practices are essential. They ensure that AI-powered service agents remain reliable, scalable, and continuously aligned with evolving customer service operations and business goals.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate vehicle creation, dealer assignment, stock validation triggers, order status updates, batch processing, scheduled reminders, and dashboard analytics. All features were verified to ensure correct functionality and performance.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized parcel and customer information in Salesforce.
- Provides better visibility into parcel booking and delivery tracking.
- Automates key logistics processes using Salesforce Flow.
- Agentforce enables conversational customer support and parcel-status assistance.
- Prompt Builder supports AI-powered responses and self-service.
- Reduces manual effort for customers, support teams, and administrators.
- Provides timely notifications and updates during the delivery process.
- Experience Cloud can provide customers with convenient access to tracking information.
- Supports integration with external logistics systems.
- Can scale to support multiple locations and growing parcel operations.

Disadvantages

- Requires accurate and up-to-date parcel and customer data.

- Dependence on Salesforce configuration and automation.
- Agentforce responses depend on properly configured prompts and available data.
- External logistics integrations may increase implementation complexity.
- Flow and automation logic require regular maintenance when business processes change.
- Users need appropriate Salesforce or Experience Cloud access to use the system.
- Initial Salesforce implementation and configuration can require significant effort.
- Expanding the system to additional locations or integrations may require further customization.

9. Conclusion:

The SwiftShipTracker demonstrates how Salesforce can be used to build a centralized parcel management and tracking solution. The project integrates custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, AgentForce AI, Prompt Builder, security controls, and reporting to support parcel booking, delivery tracking, status updates, and customer communication.

The integration of AgentForce AI with Prompt Builder and Salesforce Flow provides a conversational approach to parcel tracking, enabling customers to obtain shipment information without depending entirely on manual support. The project therefore provides a practical foundation for a scalable and user-friendly parcel management solution.

10. FUTURE SCOPE

- Extend AgentForce capabilities for more parcel-management queries.
- Enhance real-time delivery tracking.
- Expand customer self-service through Experience Cloud.
- Add more advanced delivery-performance analytics.
- Integrate additional external logistics systems.
- Improve AI prompts and conversational experiences.

- Introduce enterprise-level sandbox and CI/CD deployment.
- Expand automation for additional parcel operations.

The source identifies enterprise deployment, continuous monitoring, prompt refinement, Flow improvement, and AI-agent scalability as areas for future enhancement.

11. APPENDIX

- A. Custom Objects: Parcel, Delivery, Sender, Receiver
- B. Standard Objects: Account, Contact, Case, Task, EmailMessage, User
- C. Automation: Flows, Approval Processes, Email Alerts
- D. Apex: Apex Classes, Batch Apex
- E. AI: AgentForce AI, Prompt Builder
- F. UI: Lightning App, Lightning Record Pages, Experience Cloud
- G. Security: Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security
- H. Analytics: Reports & Dashboards
- I. Testing: Flow, AgentForce, data retrieval, security and permission validation

Thank You